CS 540 Introduction to Artificial Intelligence

Perceptron

University of Wisconsin-Madison

Spring 2025

Announcements

- **Homework:**
  - HW5 due **Monday March 3rd at 11:59 PM**
- **Midterm March 13th 7:30 – 9:00 PM**
- Class roadmap:

Machine Learning: Neural Networks I (Perceptron)
Machine Learning: Neural Networks II
Machine Learning: Neural Networks III

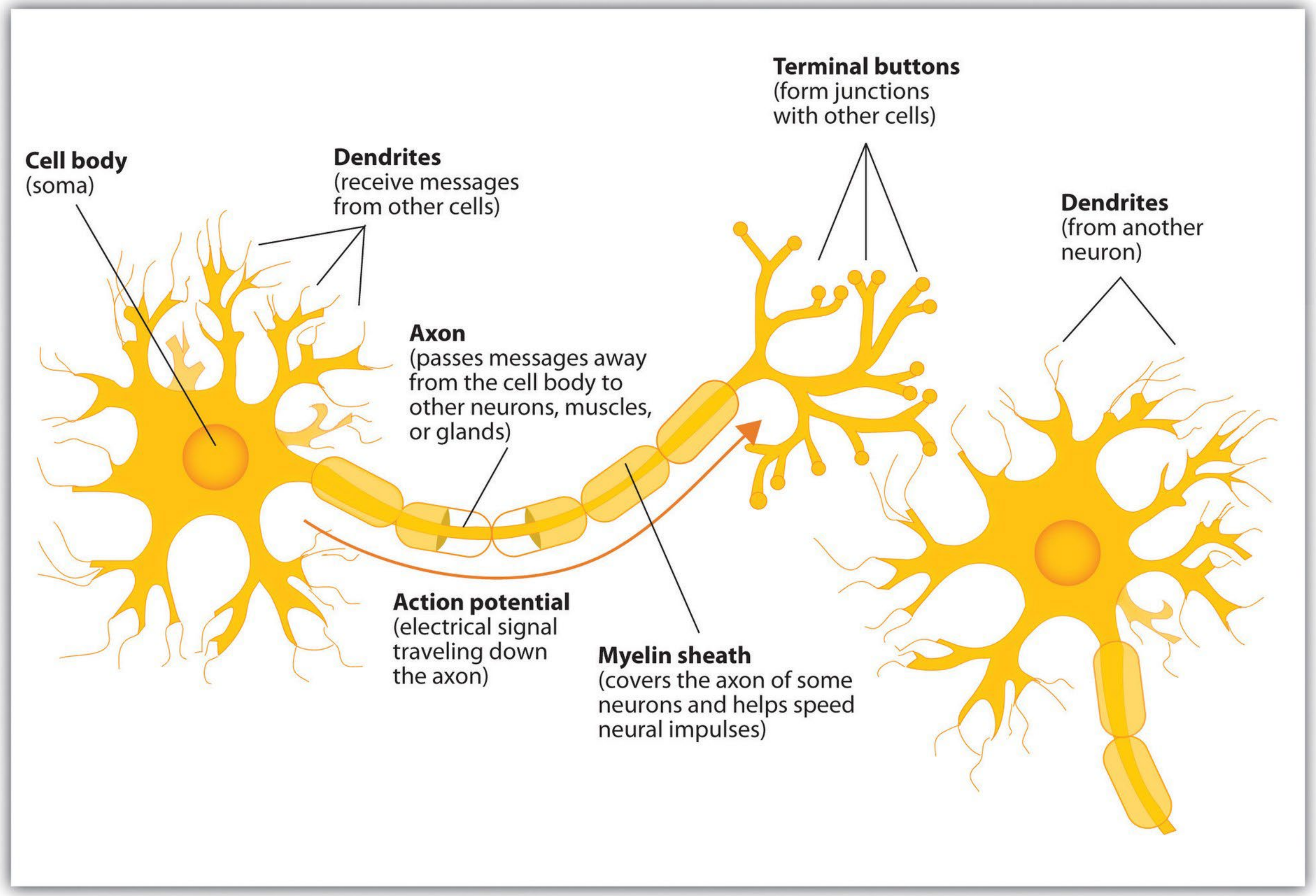
Supervised Learning

Part I: Single-layer Neural Network

Inspiration from neuroscience

- Inspirations from human brains
- Networks of simple and homogenous units

(wikipedia)

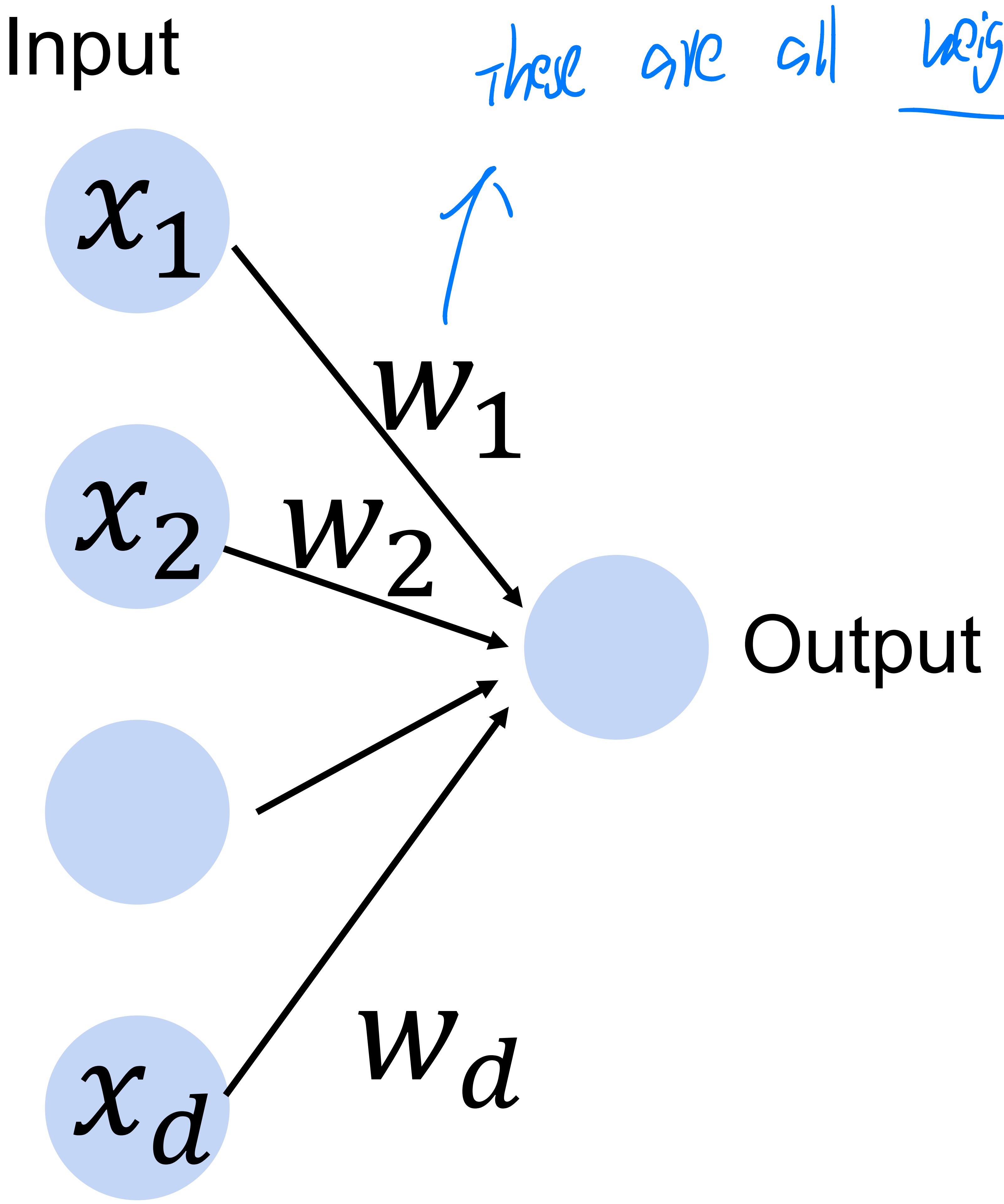


Perceptron

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} \in \mathbb{R}^d$$

Cats vs. dogs?

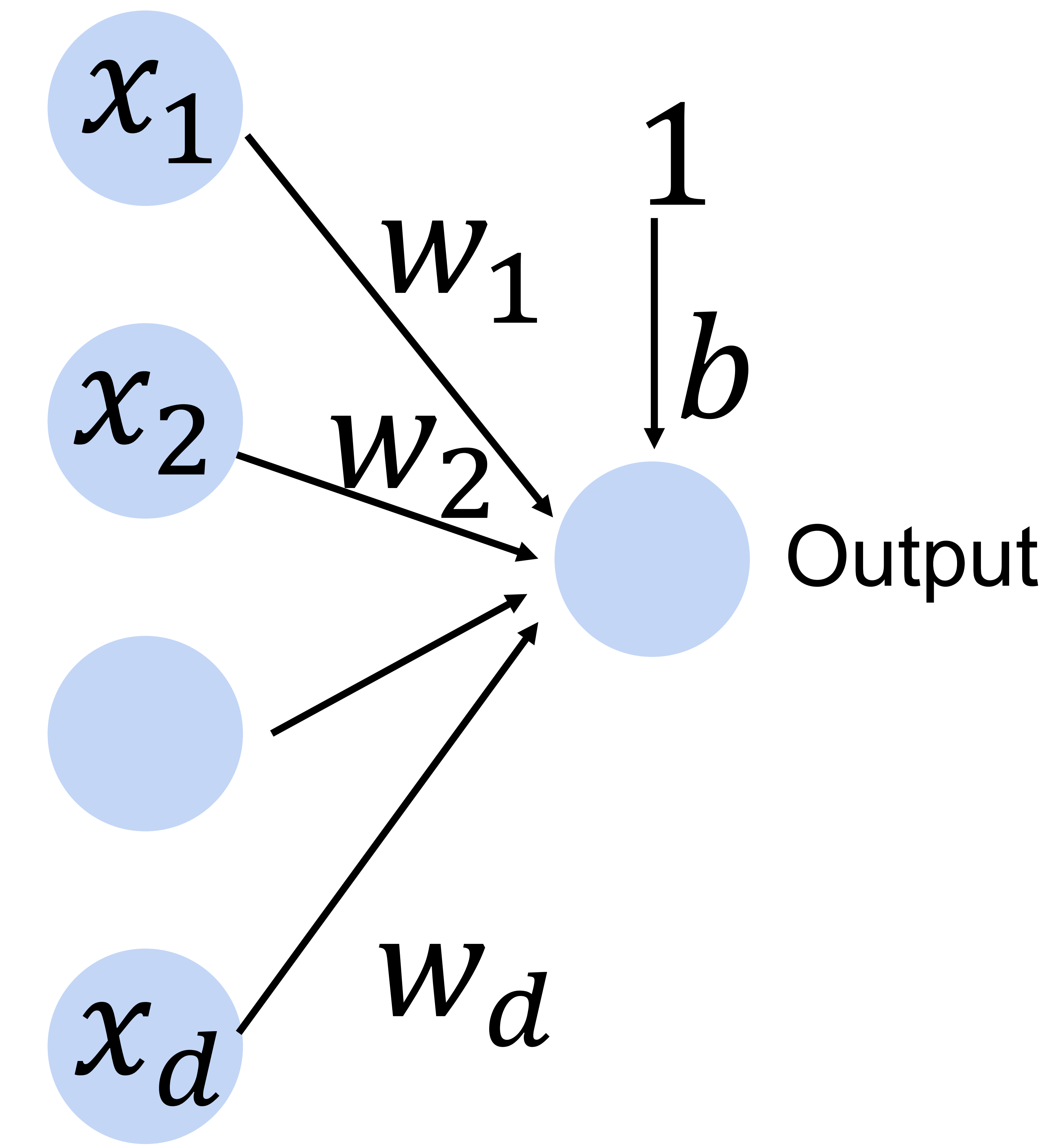
these are all weights



Linear Perceptron

- Given input $\mathbf{x}$, weight $\mathbf{w}$ and bias b, perceptron outputs:

$$f = \langle \mathbf{w}, \mathbf{x} \rangle + b$$



Cats vs. dogs?

Perceptron

(★ Remember, last slide is about linear perceptron on continuous values, here we apply a sigmoid function for classification)

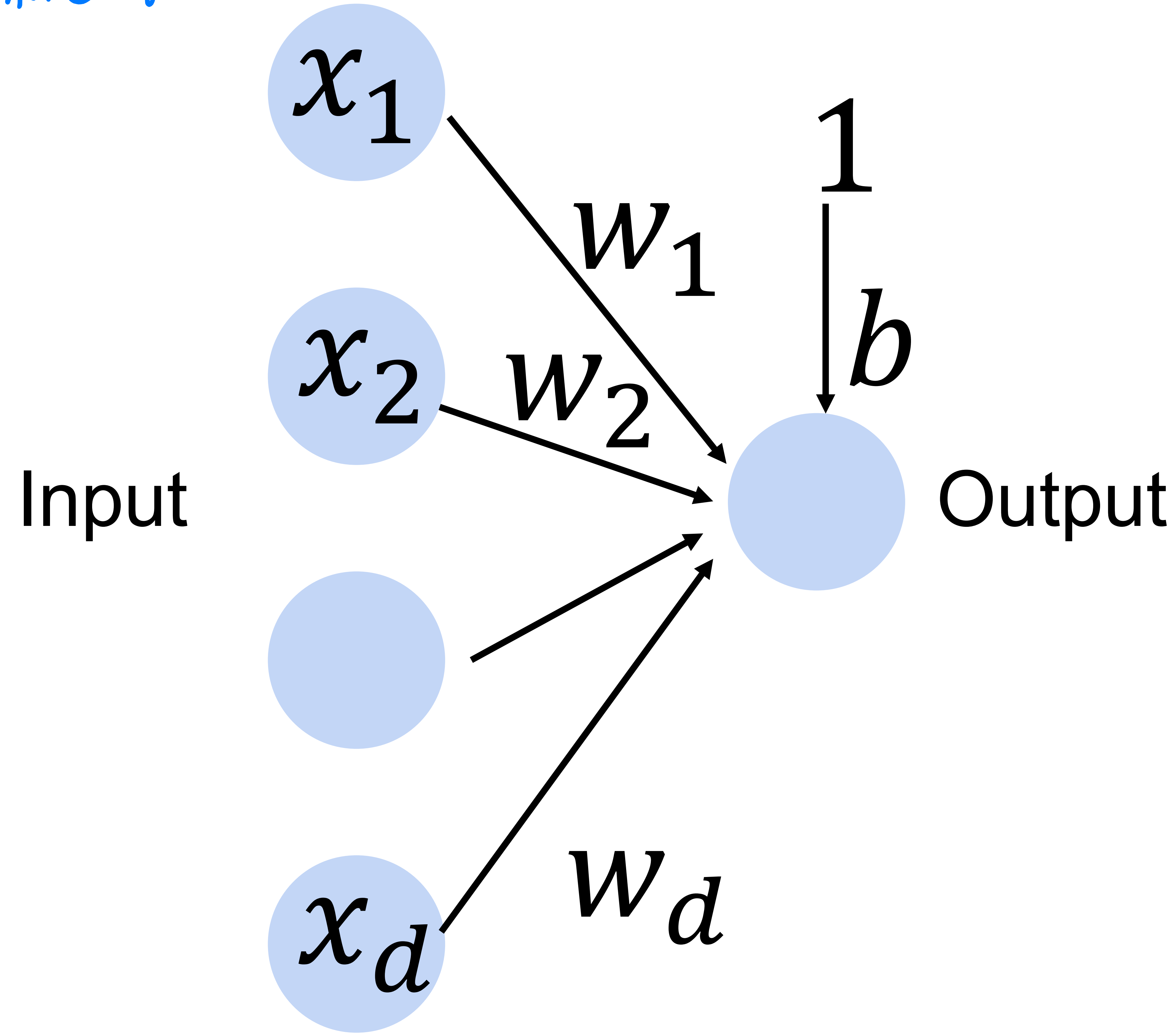
- Given input $\mathbf{x}$, weight $\mathbf{w}$ and bias b, perceptron outputs:

$$o = \sigma(\langle \mathbf{w}, \mathbf{x} \rangle + b) \qquad \sigma(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Activation function

sigmoid function

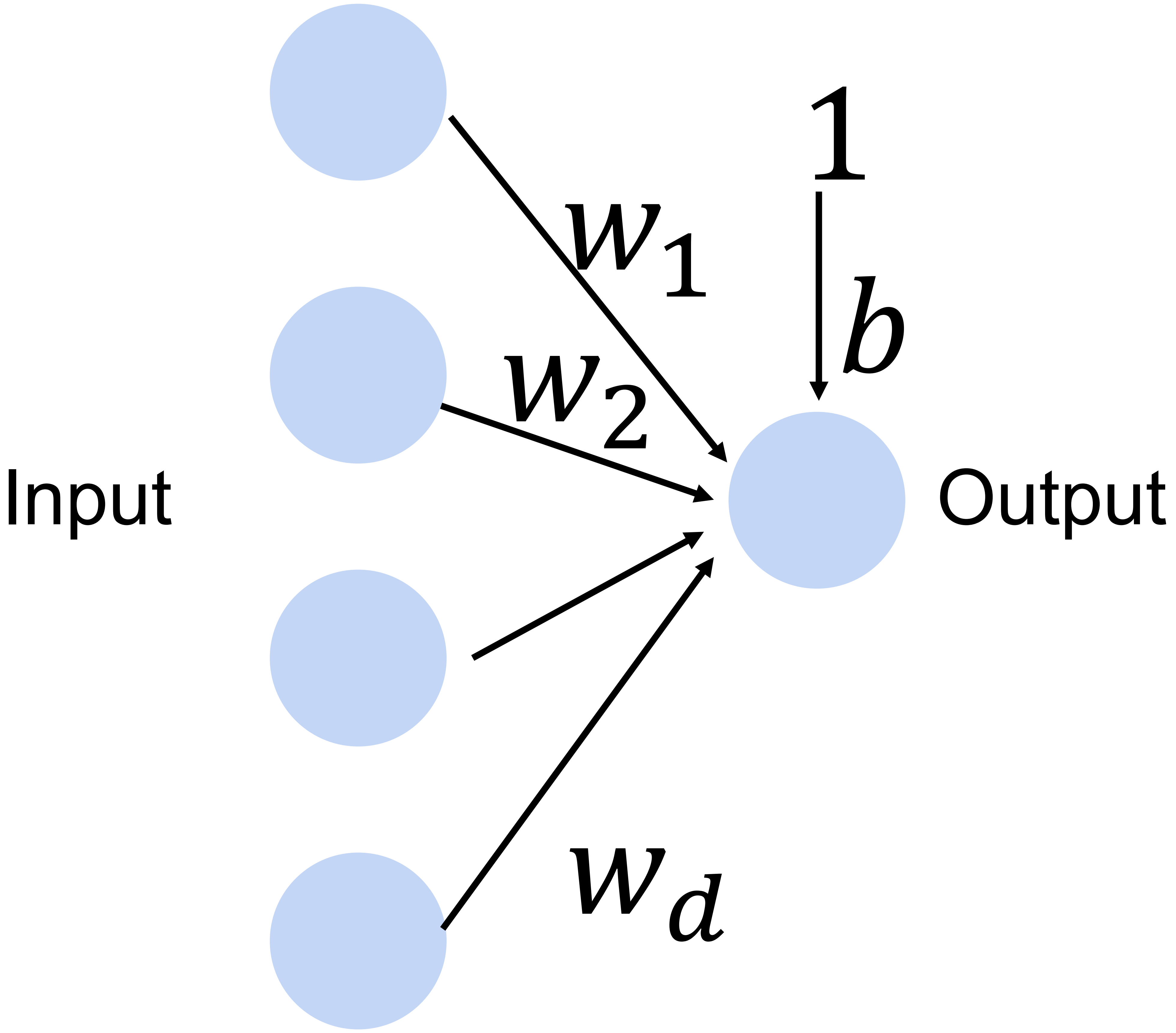
Cats vs. dogs?



Perceptron

- Goal: learn parameters $\mathbf{w} = \{w_1, w_2, \dots, w_d\}$ and b to minimize the classification error

Cats vs. dogs?



Training the Perceptron

x augmented with dimension of constant 1

$$o = \sigma(\langle \mathbf{w}, \mathbf{x} \rangle)$$

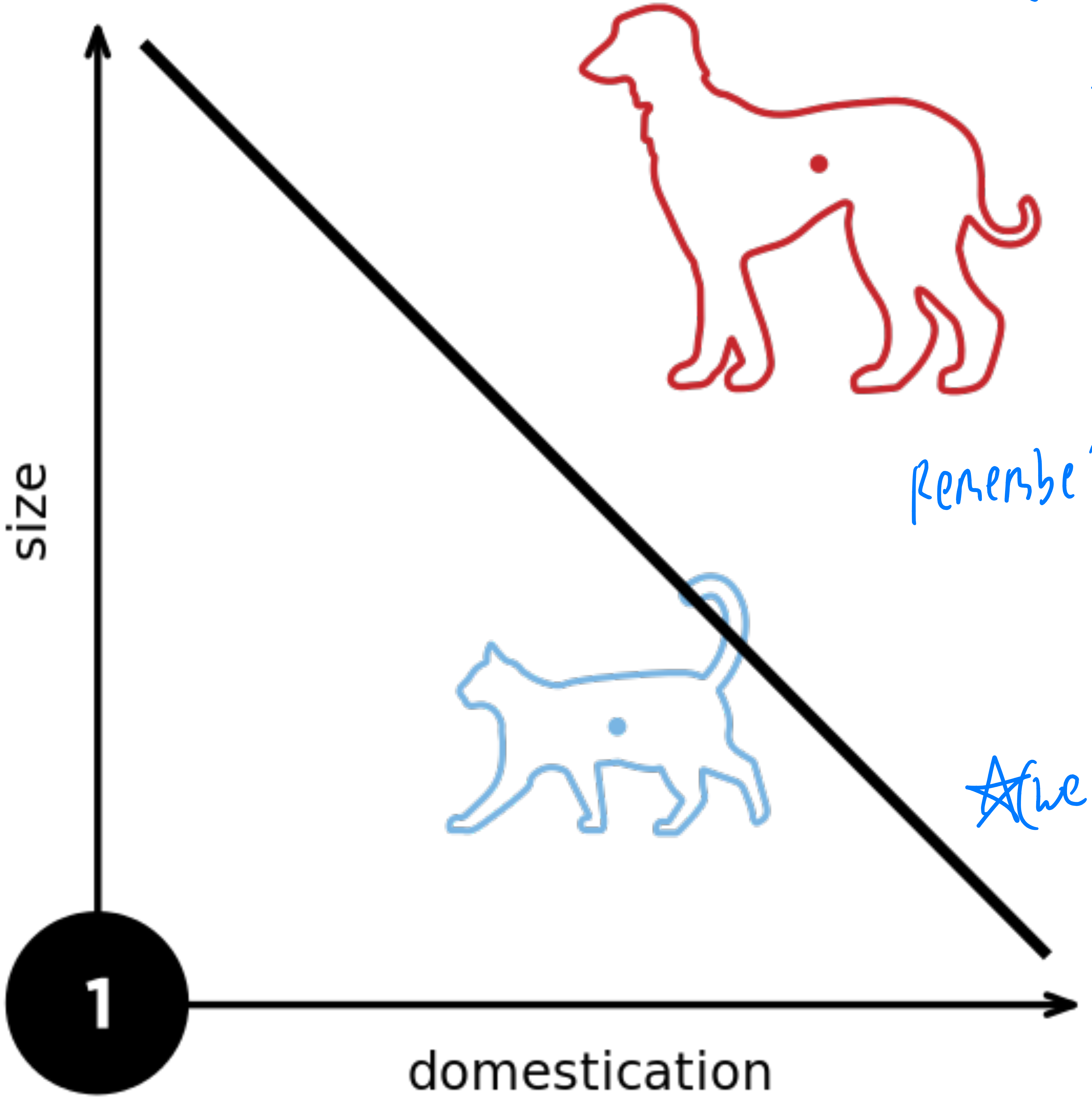
$$\sigma(x) = \begin{cases} 1 & if\, x > 0 \\ 0 & otherwise \end{cases}$$

① Run over the dataset
② check to see if making a mistake on a particular data point
③ make changes to weight if having a mistake

Perceptron Algorithm

Initialize $\vec{w} = \vec{0}$ // Initialize $\vec{w}$. $\vec{w} = \vec{0}$ misclassifies everything.
while TRUE **do** // Keep looping
  $m = 0$ // Count the number of misclassifications, m
  for $(x_i, y_i) \in D$ **do** // Loop over each (data, label) pair in the dataset, D
    if $o_i \neq y_i$ // If the pair $(\vec{x}_i, y_i)$ is misclassified
      $\vec{w} \leftarrow \vec{w} + x_i \; if \; y_i = 1, \vec{w} \leftarrow \vec{w} - x_i \;\; if \; y_i = 0$
      $m \leftarrow m + 1$ // Counter the number of misclassification
    end if
  end for
  if $m = 0$ **then** // If the most recent $\vec{w}$ gave 0 misclassifications
    break // Break out of the while-loop
  end if
end while // Otherwise, keep looping!

Perceptron



$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in R^2$$

Remember $y = \langle w, x \rangle + b$

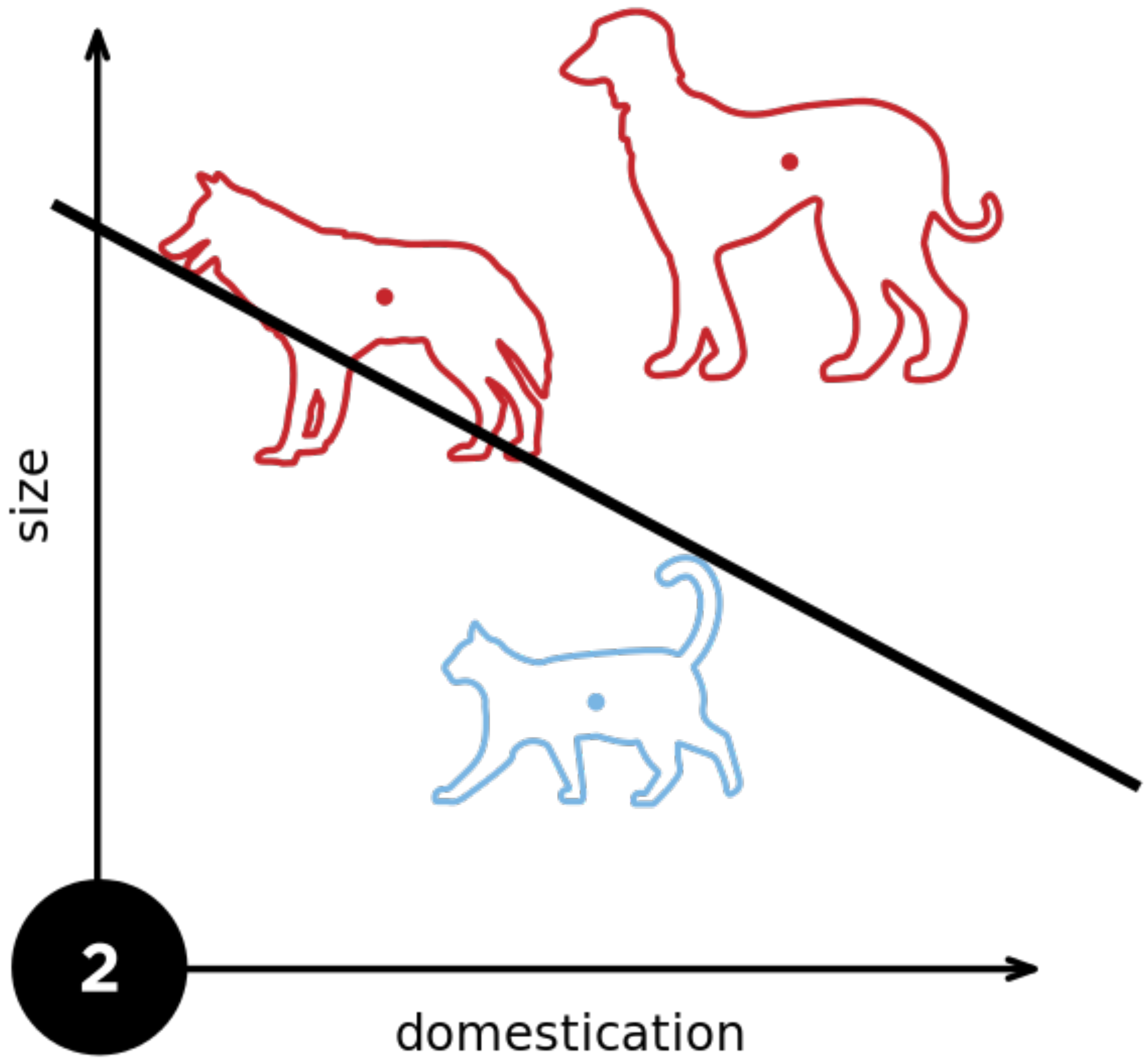
$\langle w, x \rangle$ → weights

b → b

We want to change the w/b to make it more accurate. And that's it)

From wikipedia

Perceptron



From wikipedia

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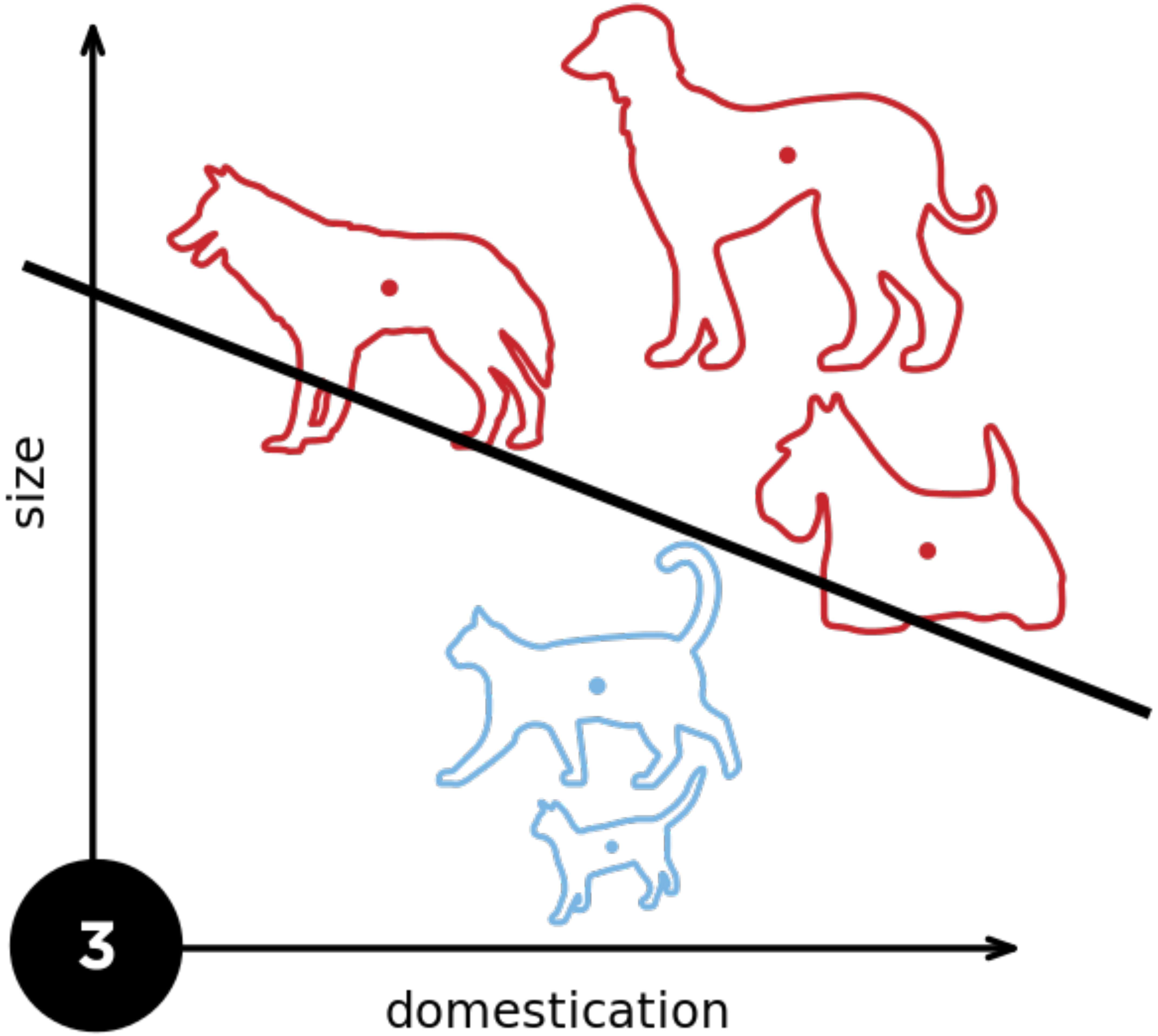
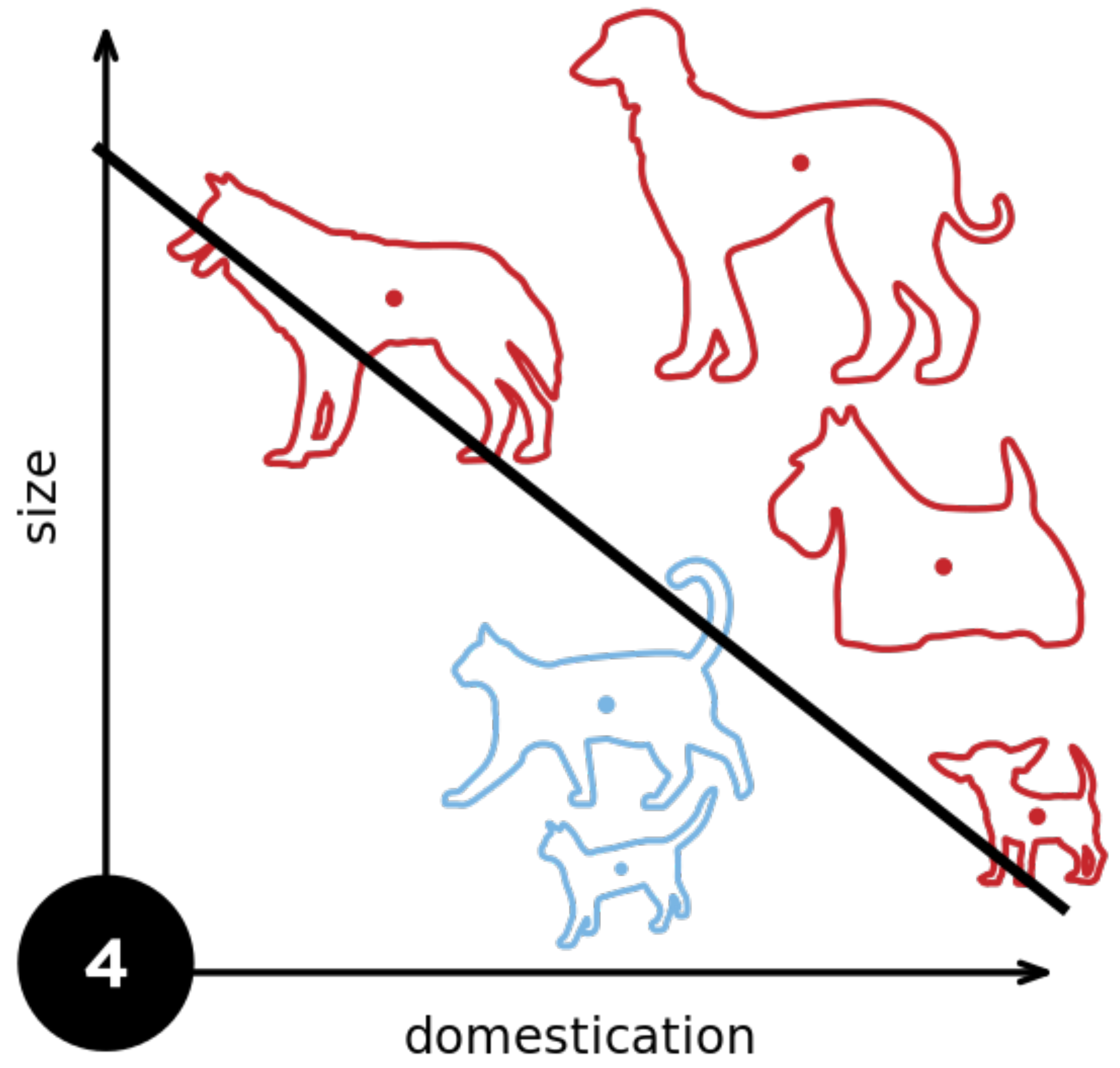


From wikipedia

Perceptron



From wikipedia

Learning AND function using perceptron

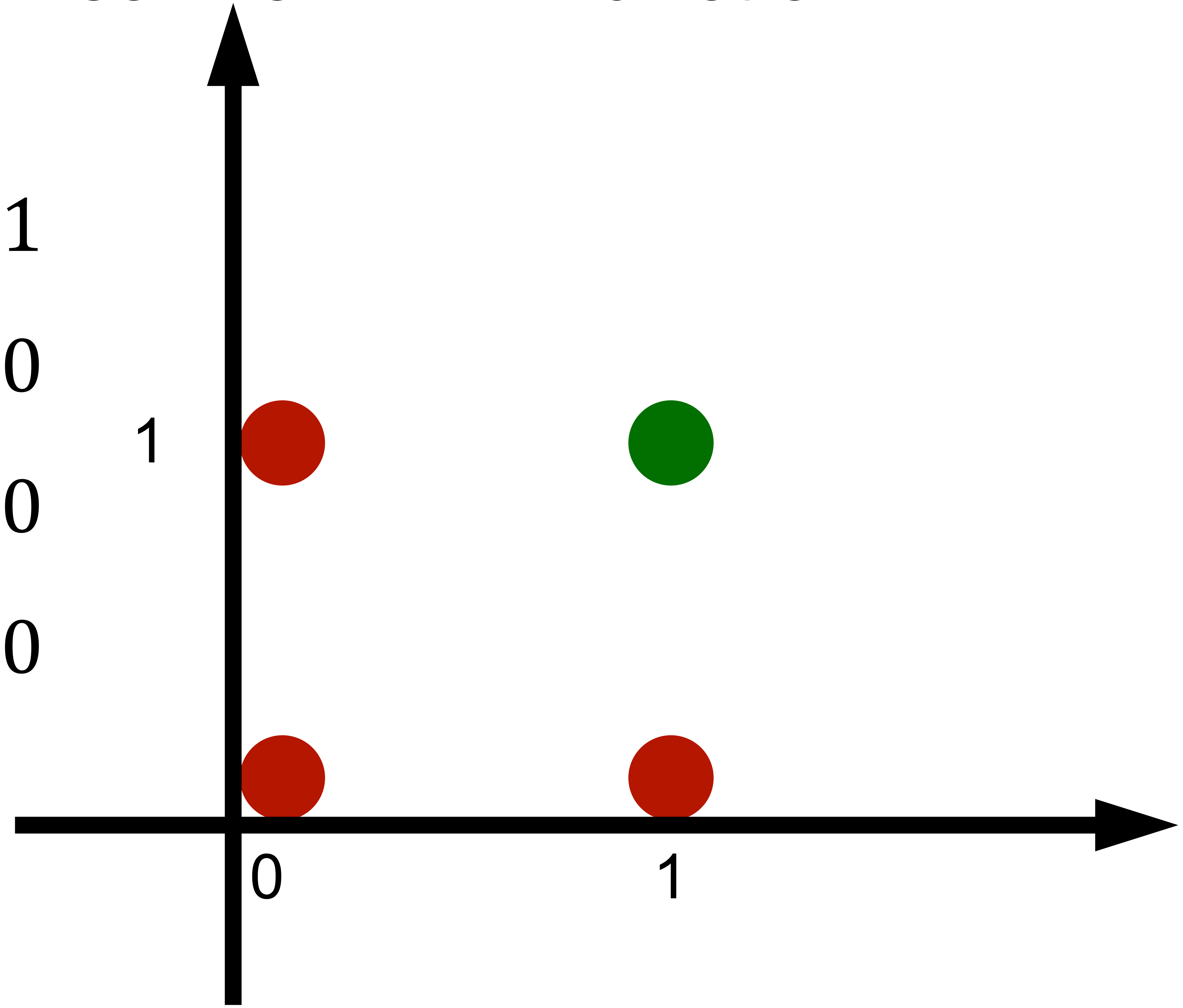
The perceptron can learn an AND function

$x_1 = 1, x_2 = 1, y = 1$

$x_1 = 1, x_2 = 0, y = 0$

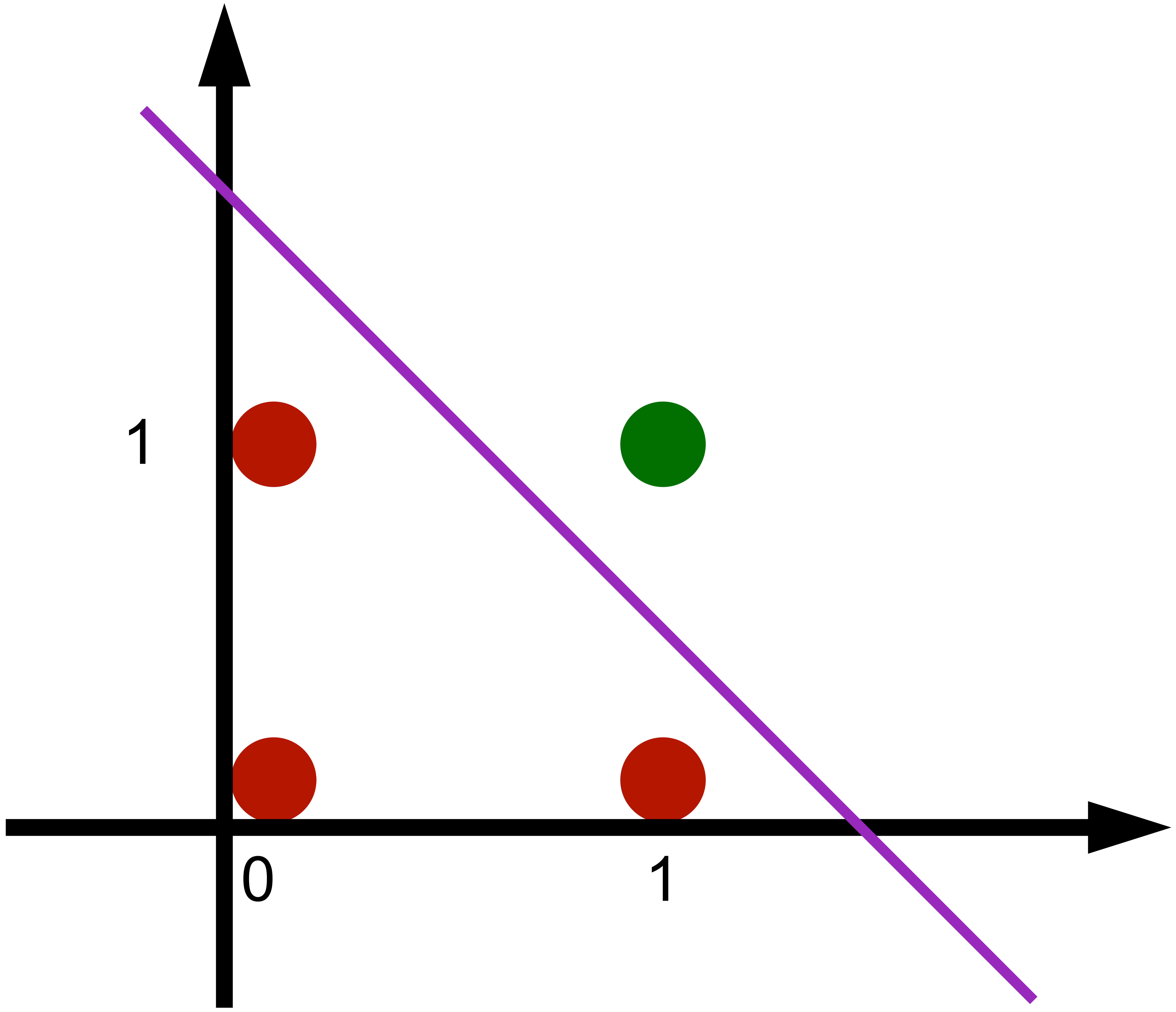
$x_1 = 0, x_2 = 1, y = 0$

$x_1 = 0, x_2 = 0, y = 0$



Learning AND function using perceptron

The perceptron can learn an AND function



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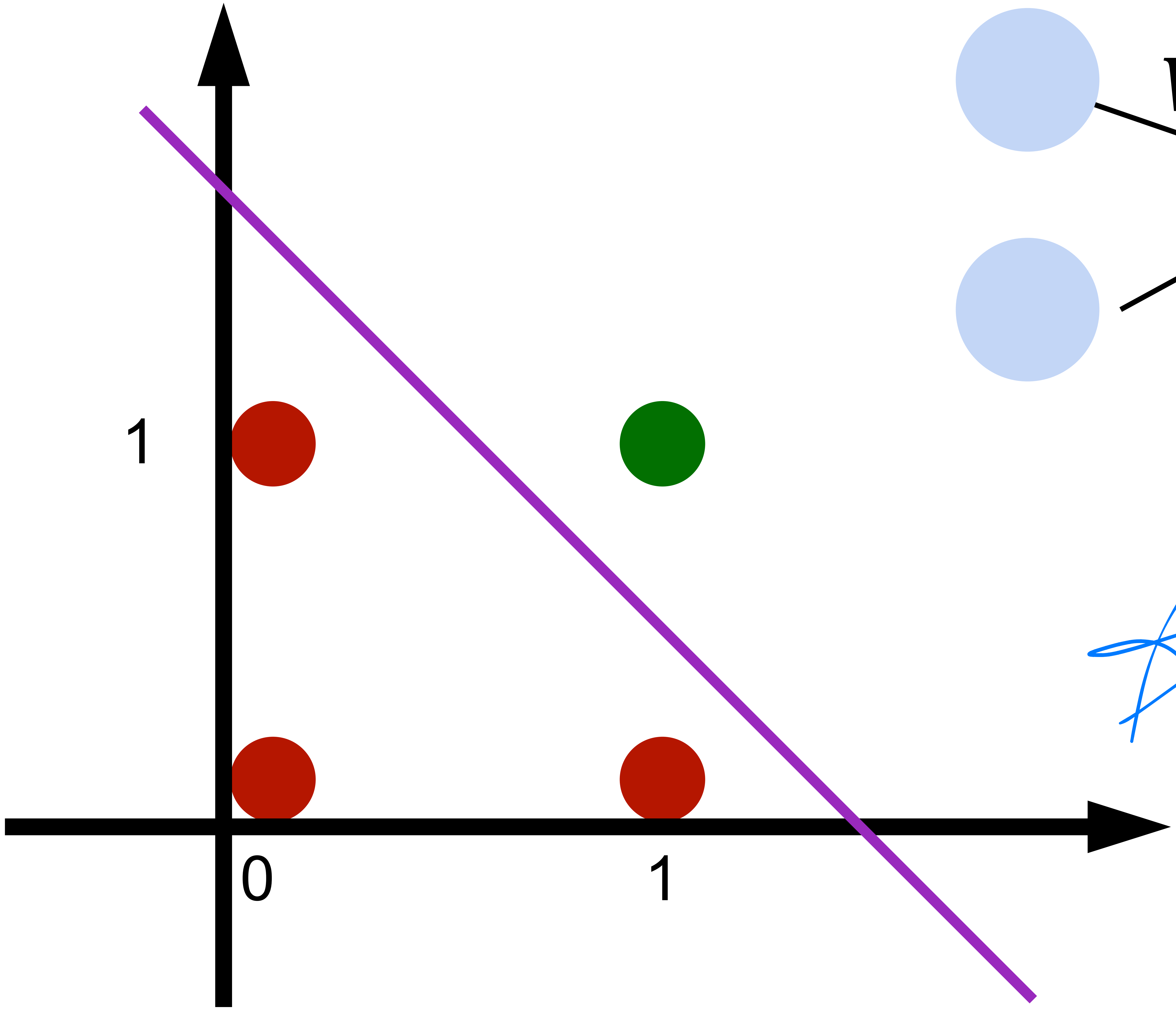
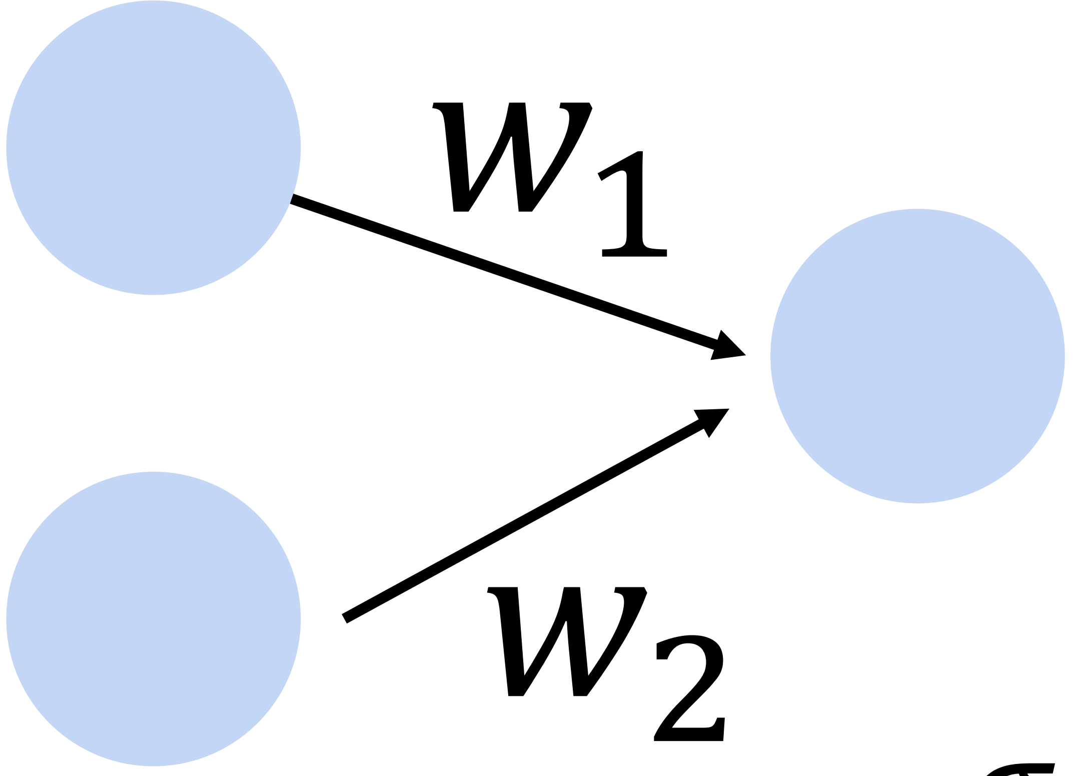




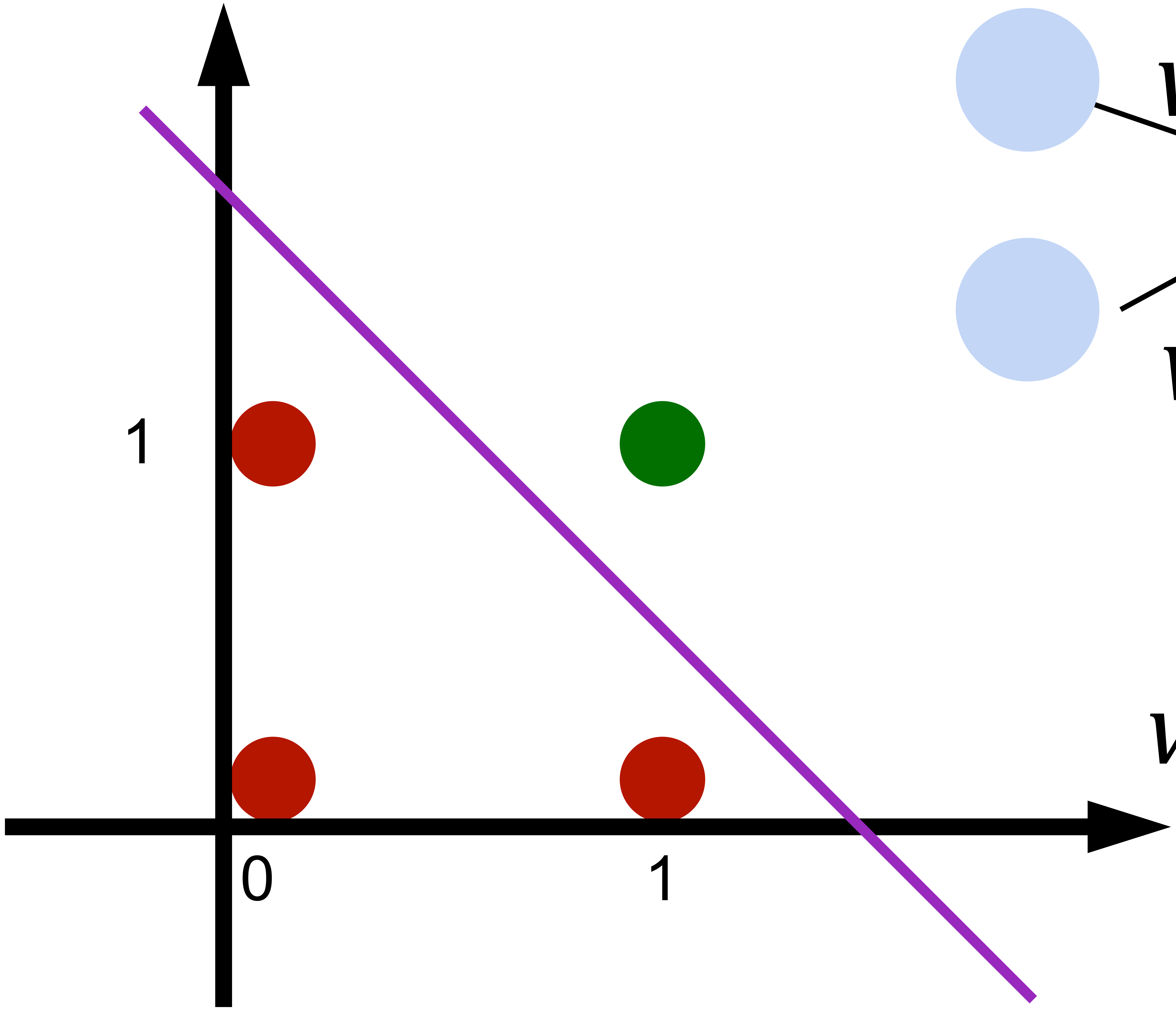
Output $\sigma(x_1 w_1 + x_2 w_2 + \underbrace{b}_{\text{bias}})$

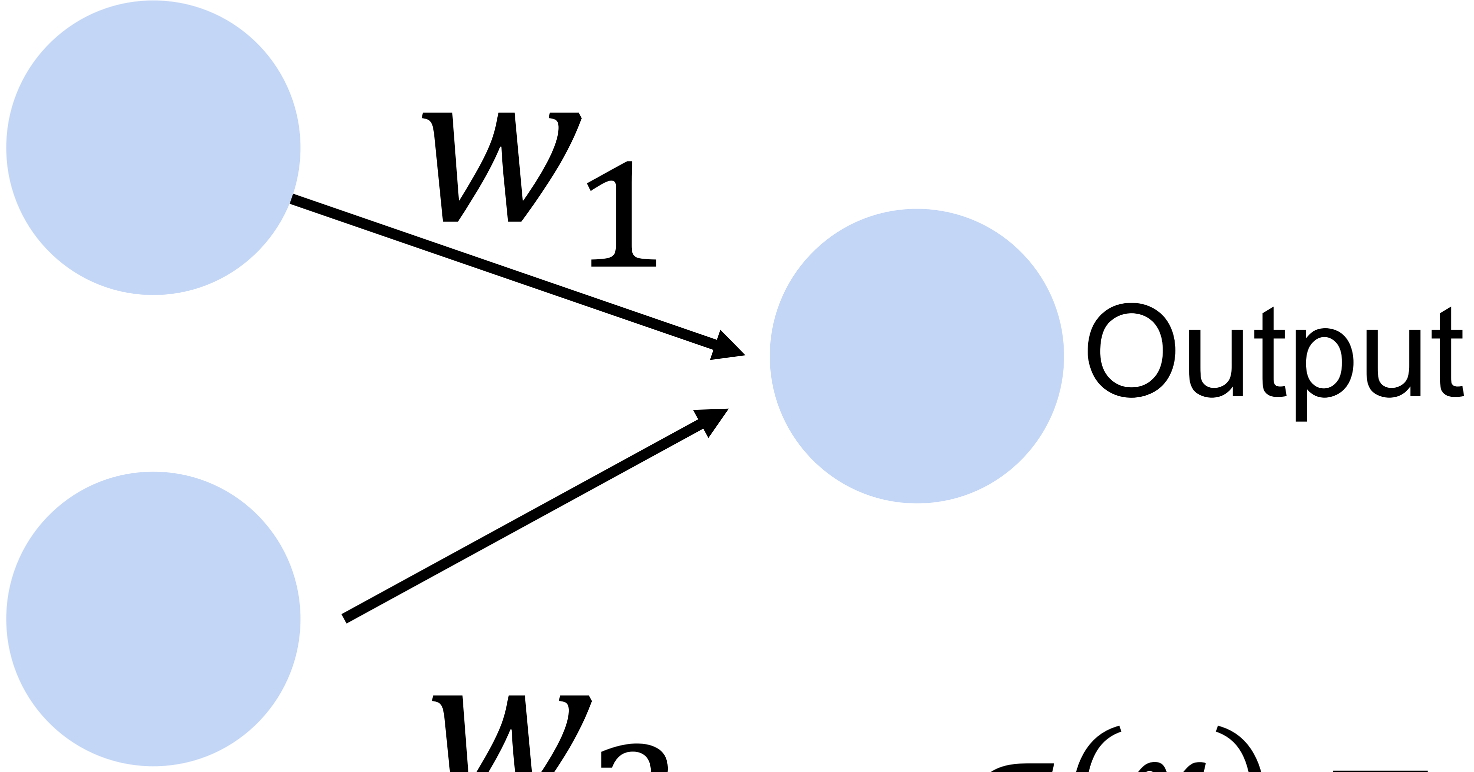
$$\sigma(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

What's w and b?

Learning AND function using perceptron

The perceptron can learn an AND function





$\sigma(x_1 w_1 + x_2 w_2 + b)$

$$\sigma(x) = \begin{cases} 1 & if\ x > 0 \\ 0 & otherwise \end{cases}$$

$$w_1 = 1, w_2 = 1, b = -1.5$$

Learning OR function using perceptron

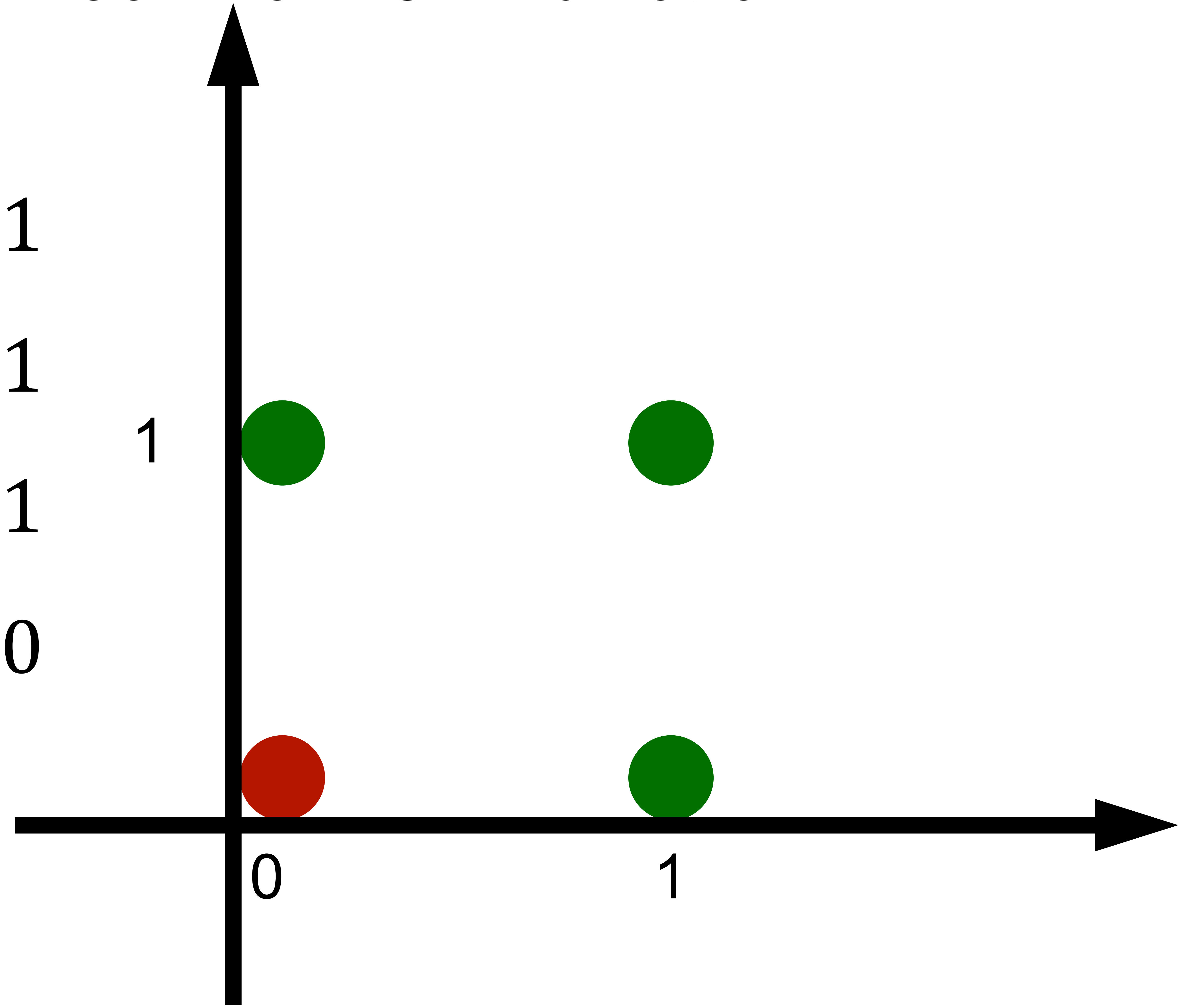
The perceptron can learn an OR function

$x_1 = 1, x_2 = 1, y = 1$

$x_1 = 1, x_2 = 0, y = 1$

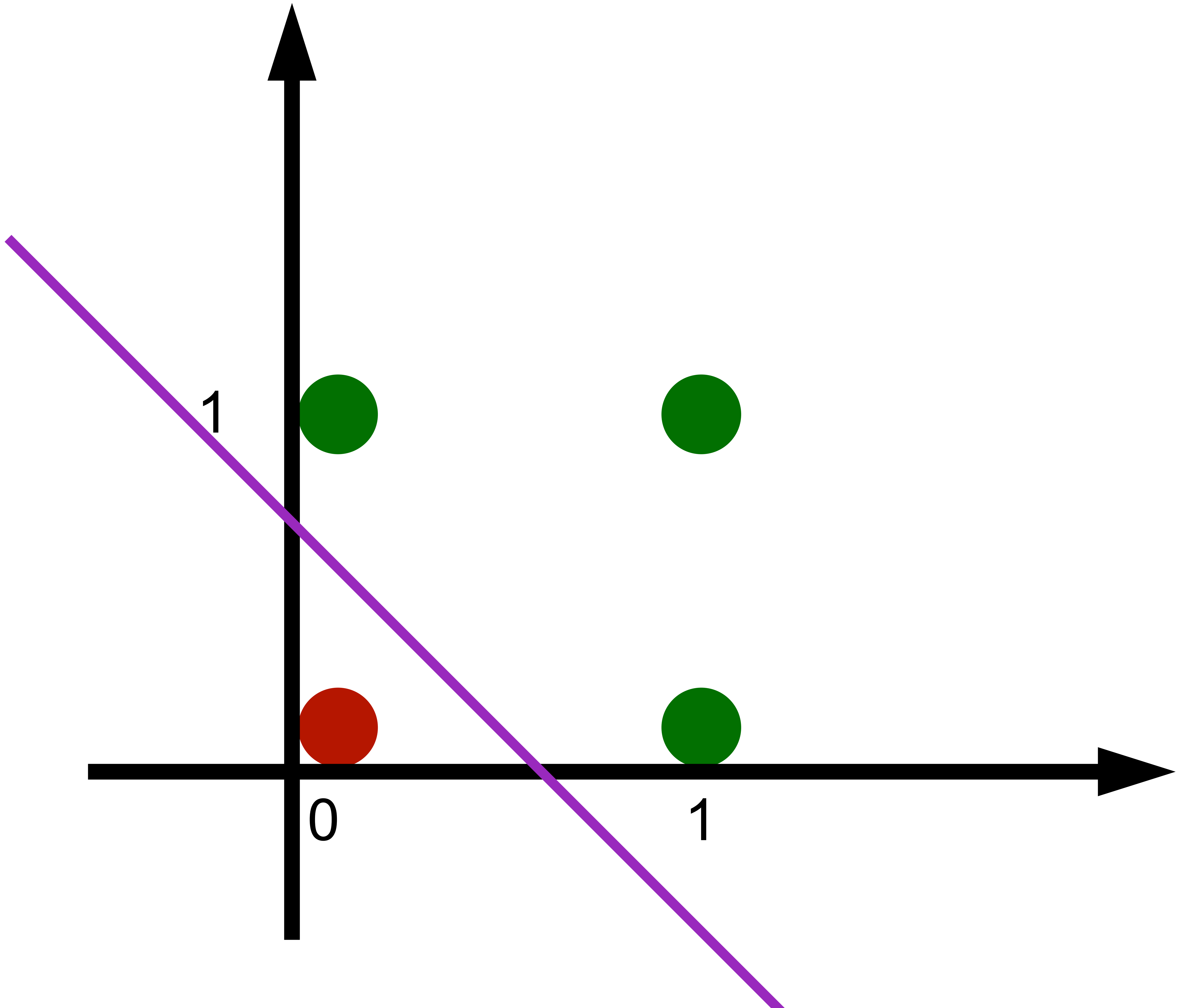
$x_1 = 0, x_2 = 1, y = 1$

$x_1 = 0, x_2 = 0, y = 0$



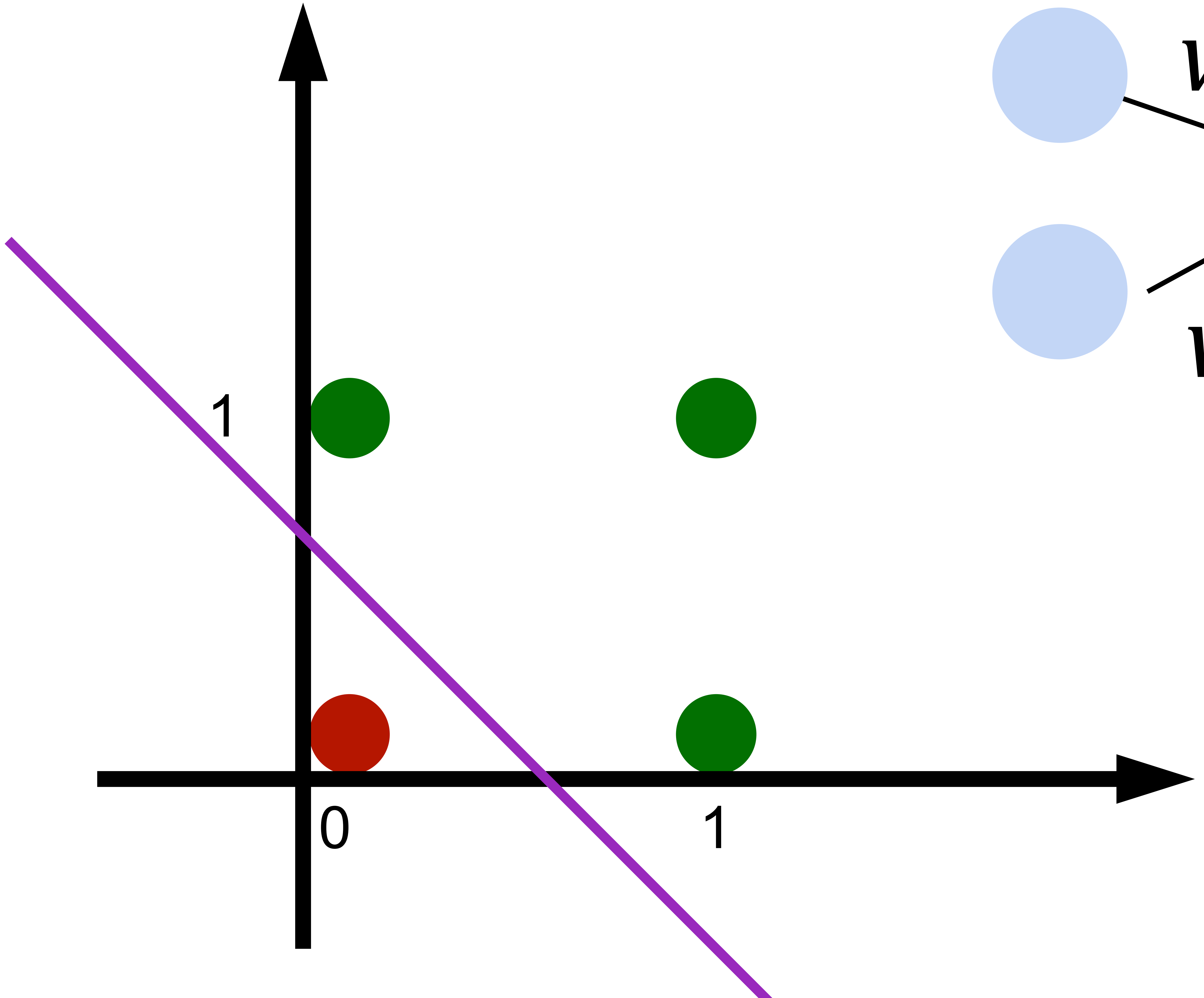
Learning OR function using perceptron

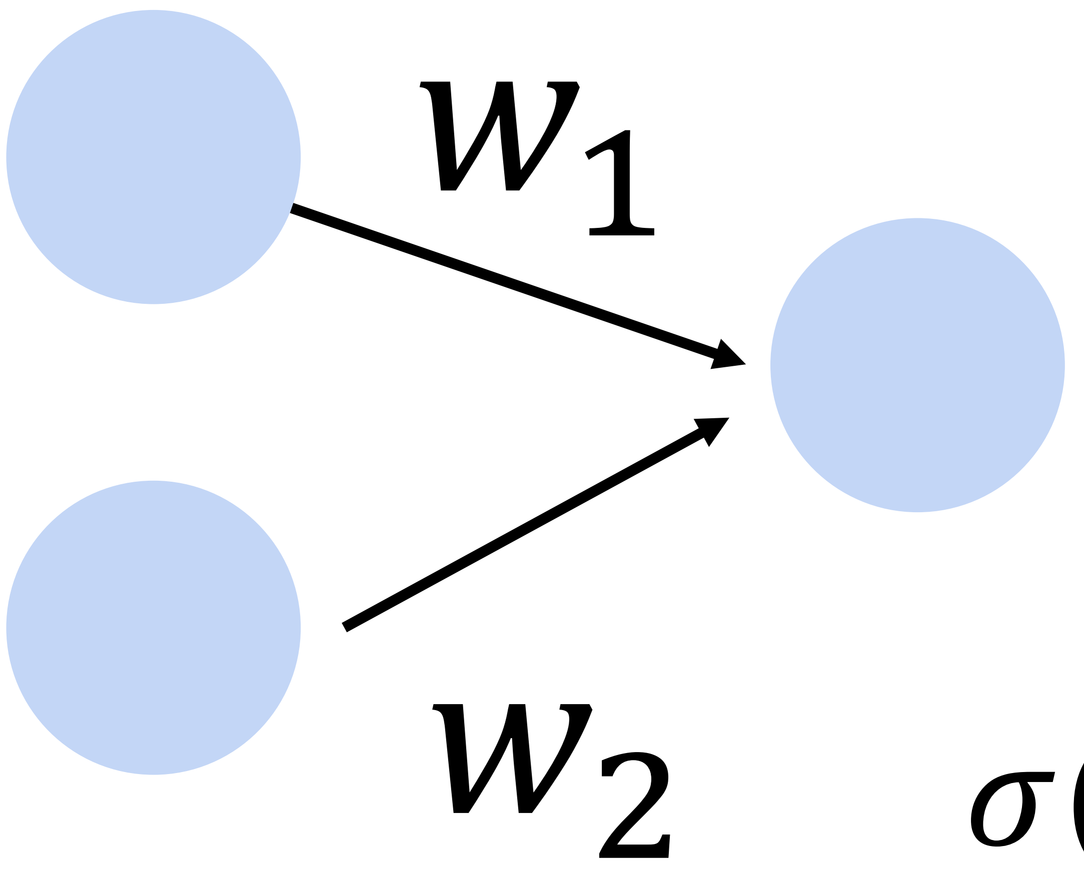
The perceptron can learn an OR function



Learning OR function using perceptron

The perceptron can learn an OR function





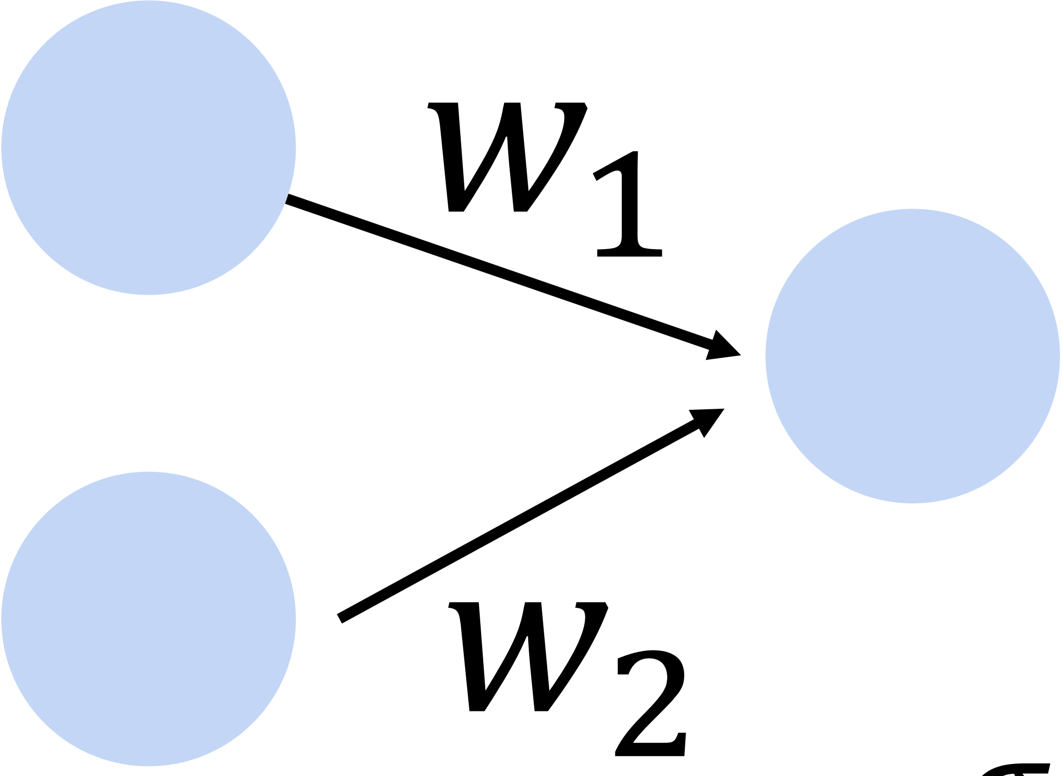
Output $\sigma(x_1 w_1 + x_2 w_2 + b)$

$$\sigma(x) = \begin{cases} 1 & if\ x > 0 \\ 0 & otherwise \end{cases}$$

What's w and b?

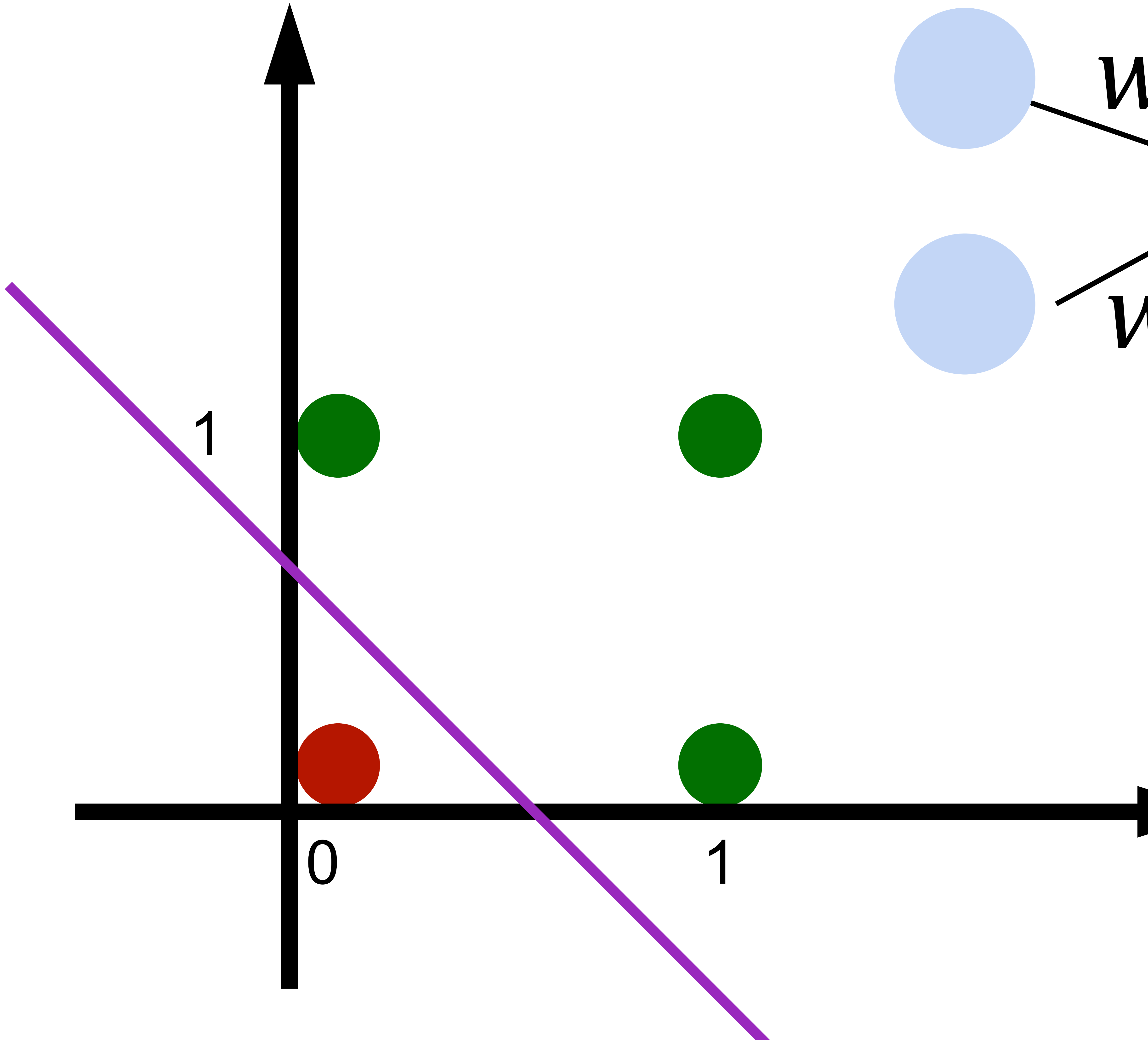
Learning OR function using perceptron

The perceptron can learn an OR function



Output $\sigma(x_1 w_1 + x_2 w_2 + b)$

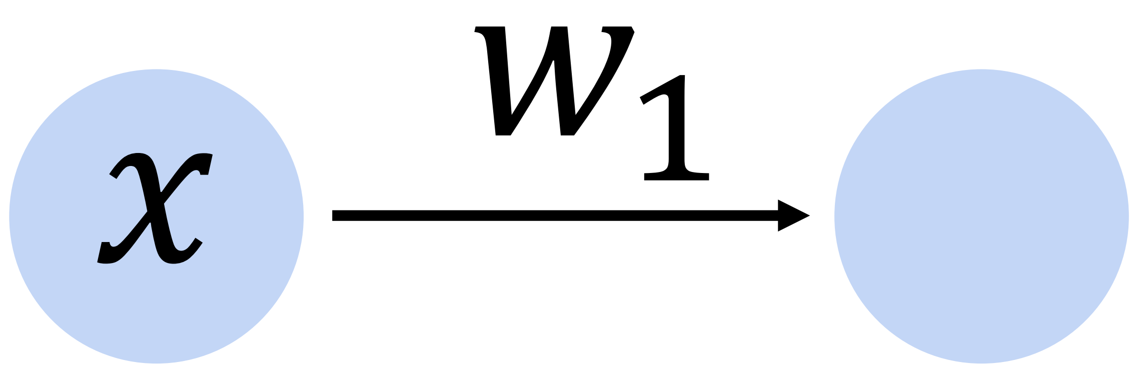
$$\sigma(x) = \begin{cases} 1 & if x > 0 \\ 0 & otherwise \end{cases}$$



$$w_1 = 1, w_2 = 1, b = -0.5$$

Learning NOT function using perceptron

The perceptron can learn NOT function (single input)



Output $\sigma(xw_1 + b)$

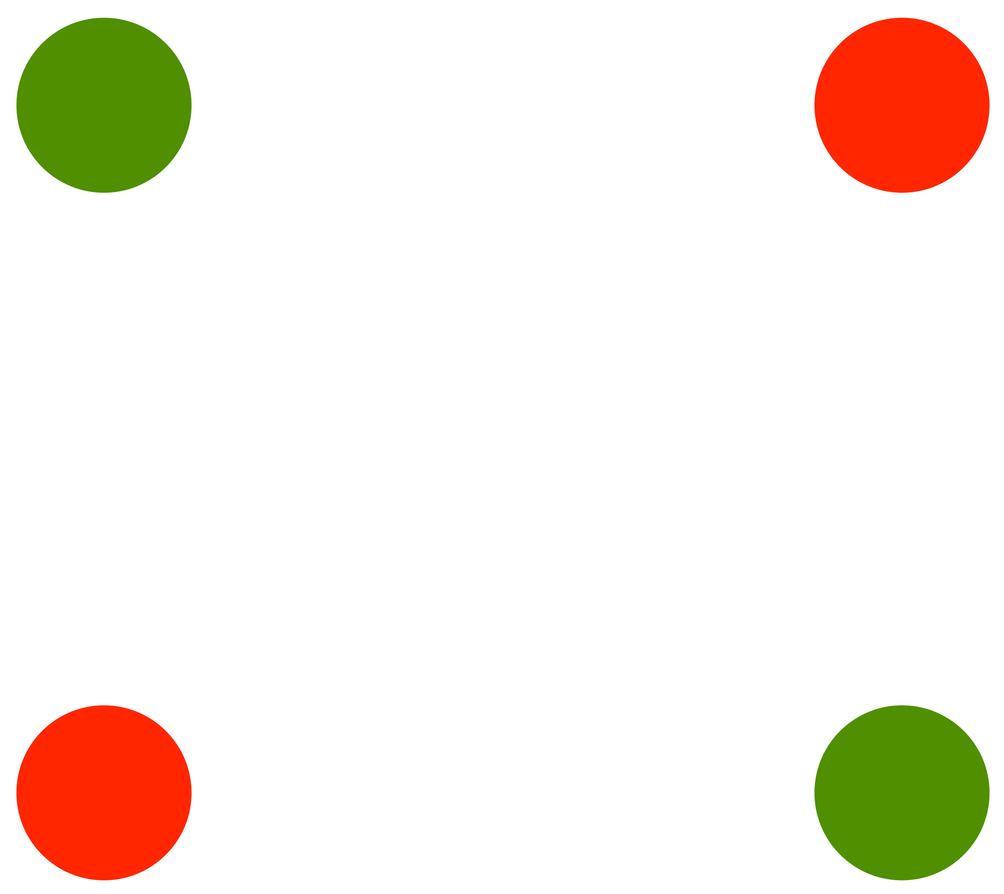
$$\sigma(x) = \begin{cases} 1 & if\ x > 0 \\ 0 & otherwise \end{cases}$$

$$w_1 = -1, b = 0.5$$

XOR Problem (Minsky & Papert, 1969)

The perceptron cannot learn an XOR function
(it can only generate linear separators)

(Not doable in linear perceptron)



This contributed to the first AI winter

Quiz Break

Consider the linear perceptron with x as the input. Which function can the linear perceptron compute?

(1) $y = \text{ax} + \text{b}$

(2) $y = \text{ax}^2 + bx + c$

A. (1)

B. (2)

C. (1)(2)

D. None of the above

Quiz Break

Consider the linear perceptron with x as the input. Which function can the linear perceptron compute?

(1) $y = \mathrm{ax} + b$

(2) $y = \mathrm{ax}^2 + bx + c$

A. (1)

B. (2)

C. (1)(2)

D. None of the above

Answer: A. All units in a linear perceptron are linear.
Thus, the model can not present non-linear functions.

Quiz Break

Perceptron can be used for representing:

A. AND function
B. OR function
C. XOR function
D. Both AND and OR function

Quiz Break

Perceptron can be used for representing:

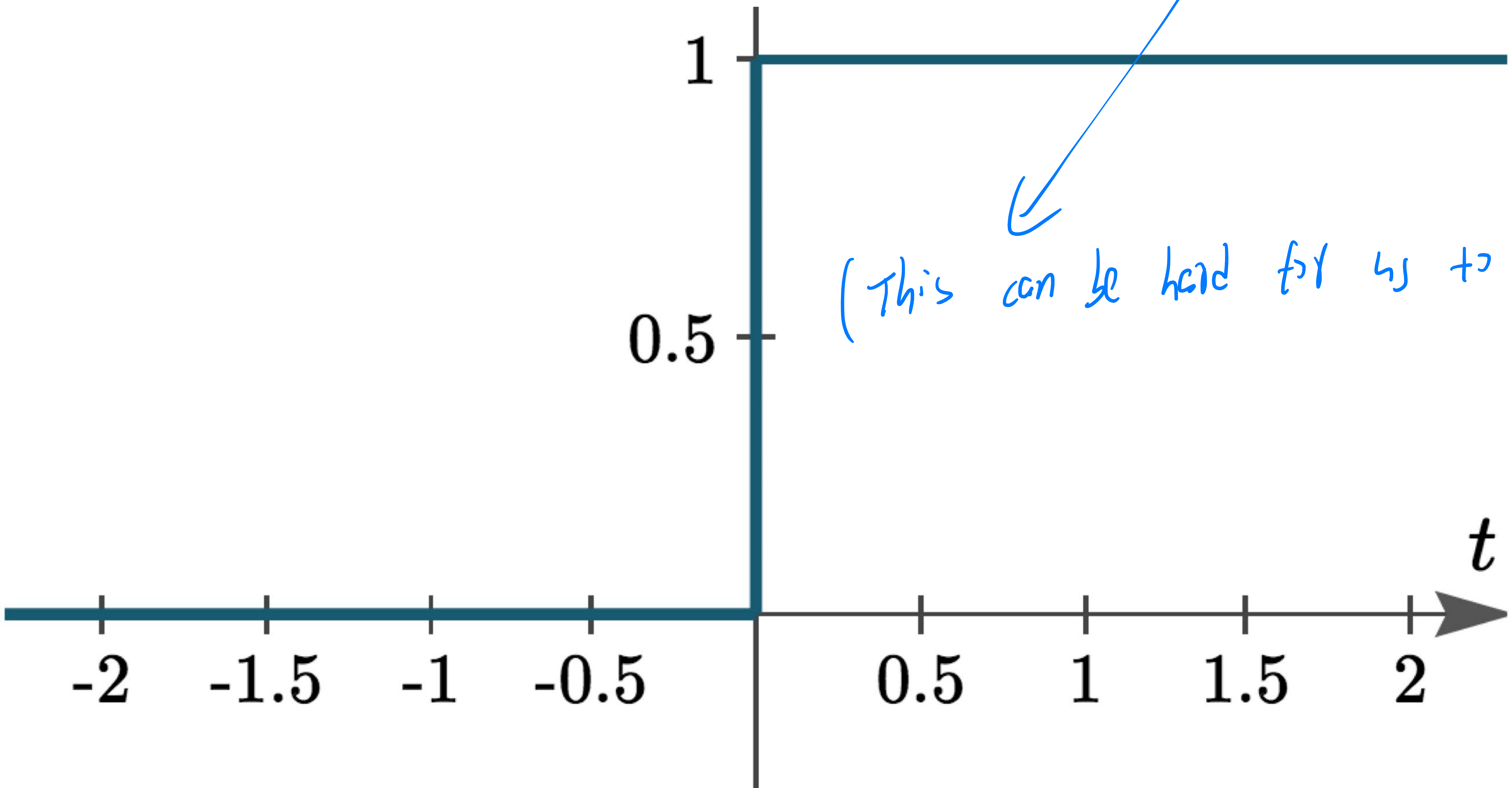
A. AND function
B. OR function
C. XOR function
D. Both AND and OR function

Step Function activation

Step function is discontinuous, which cannot be used for gradient descent

$$\sigma(x) = \begin{cases} 1 & if\ x > 0 \\ 0 & otherwise \end{cases}$$

(This can be hard for us to optimize in general)

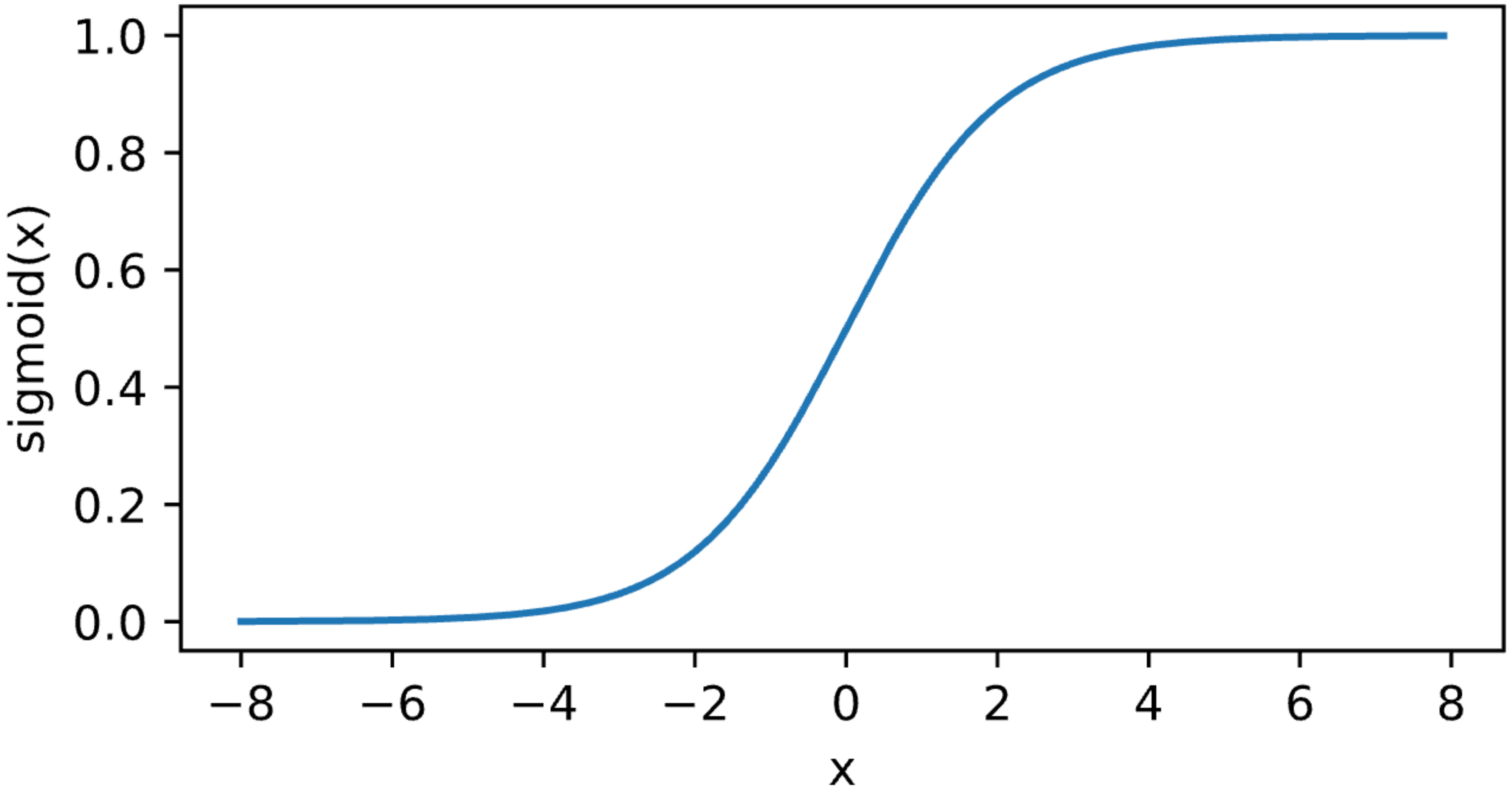


Sigmoid/Logistic Activation

Map input into [0, 1], a **soft** version of $\sigma(z) = \begin{cases} 1 & if z > 0 \\ 0 & otherwise \end{cases}$

(Improved version of the previous activation function)

$$\sigma(z) = \mathrm{sigmoid}(z) = \frac{1}{1 + \exp(-z)}$$

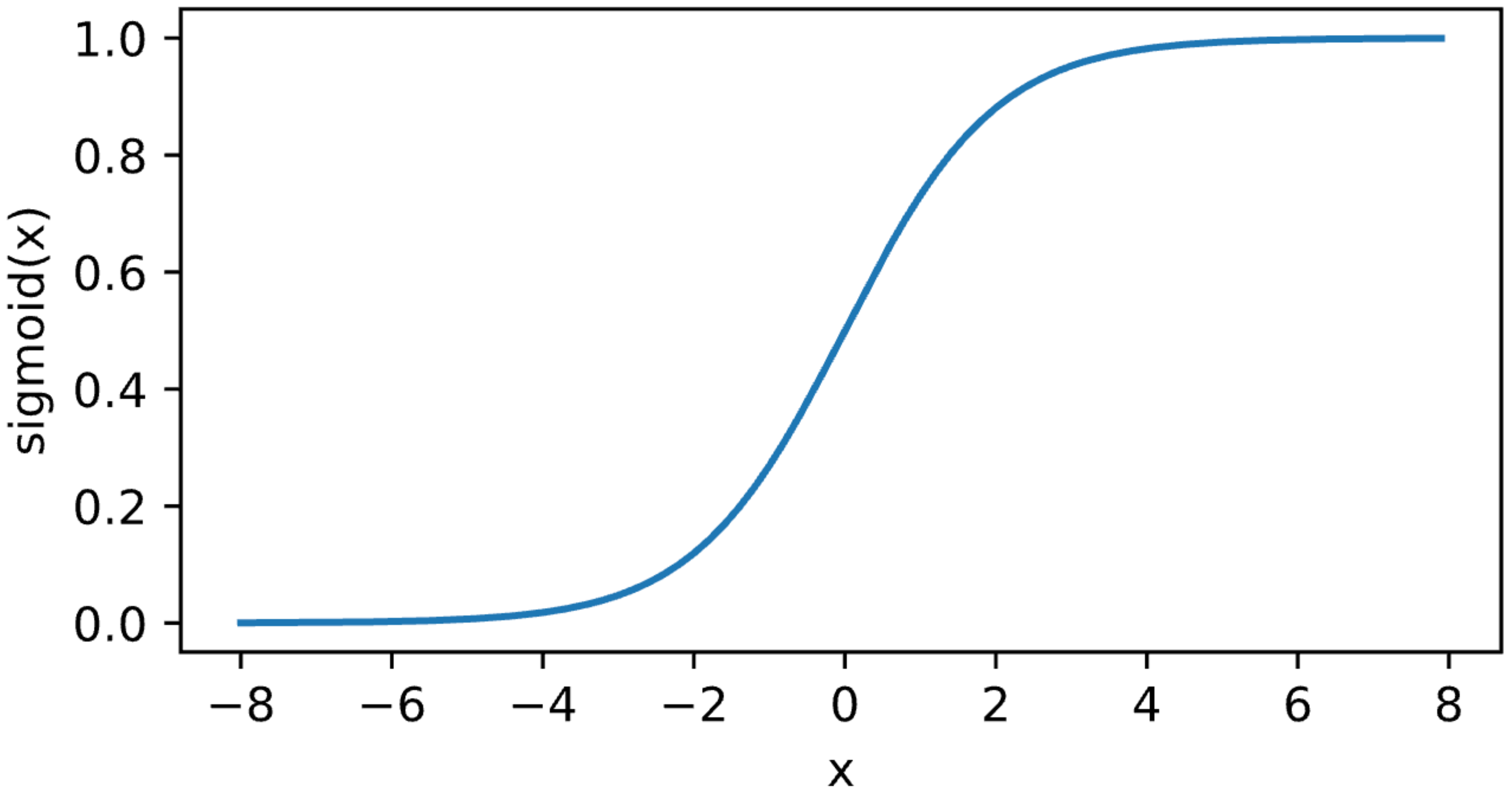


Logistic regression

$\mathbf{x} \in \mathbb{R}^d, y = \{-1, +1\}$

$$p(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^T\mathbf{x})}$$

$$p(y = -1|\mathbf{x}) = 1 - \sigma(\mathbf{w}^T\mathbf{x}) = \frac{1}{1 + \exp(\mathbf{w}^T\mathbf{x})}$$



Logistic regression

Given: $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ $\quad$ $\mathbf{x} \in \mathbb{R}^d, y = \{-1, +1\}$

Training: maximize likelihood estimate (on the conditional probability)

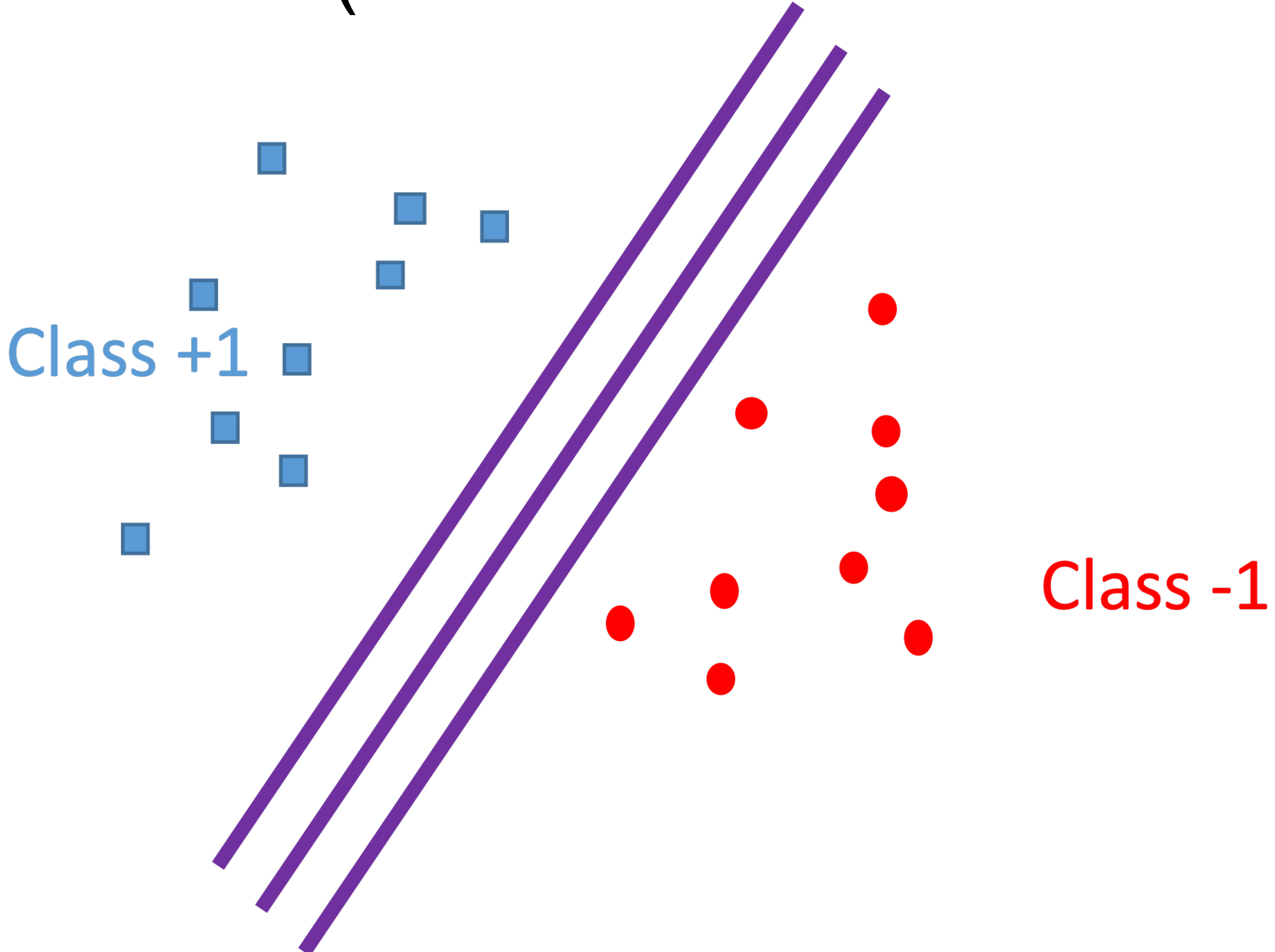
$$\max_{\mathbf{w}} \sum_i \log \frac{1}{1 + \exp(-y_i \mathbf{w}^T \mathbf{x}_i)}$$

Logistic regression

Given: $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ $\quad \mathbf{x} \in \mathbb{R}^d, y = \{-1, +1\}$

Training: maximize likelihood estimate (on the conditional probability)

When training data is linearly separable, many solutions



Logistic regression

Given: $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ $\quad \mathbf{x} \in \mathbb{R}^d, y = \{-1, +1\}$

Regularization

Training: maximum A posteriori (MAP)

$$\min_{\mathbf{w}} \sum_i -\log \frac{1}{1+\exp(-y_i \mathbf{w}^T \mathbf{x}_i)} + \frac{\lambda}{2} \| \mathbf{w} \|_2^2$$

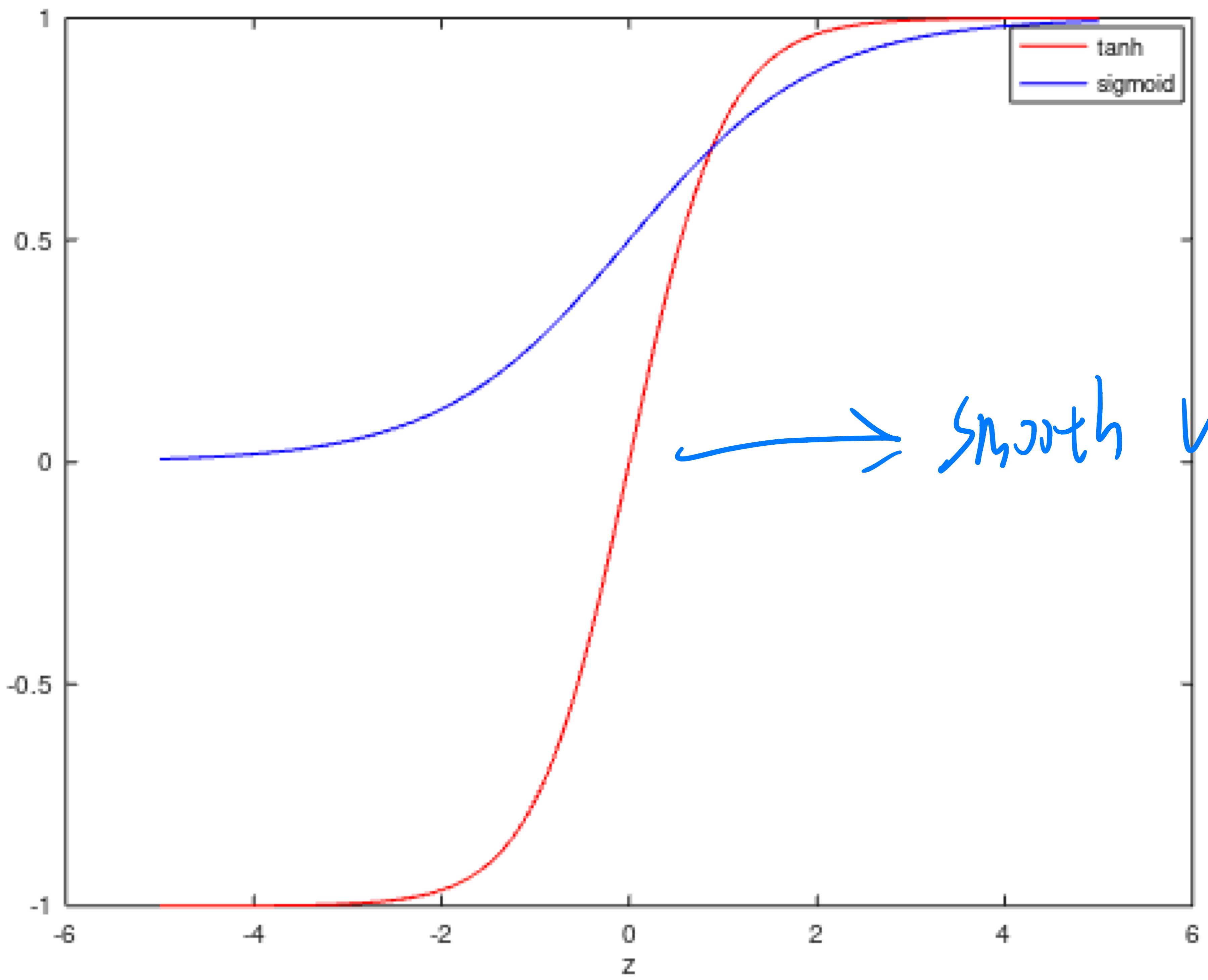
- Convex optimization
- Solve via (stochastic) gradient descent

Tanh Activation

Map inputs into (-1, 1)

$$\sigma(z) = tanh(z) = \frac{1 - \exp(-2z)}{1 + \exp(-2z)}$$

$$tanh(z) = 2sigmoid(2z) - 1$$

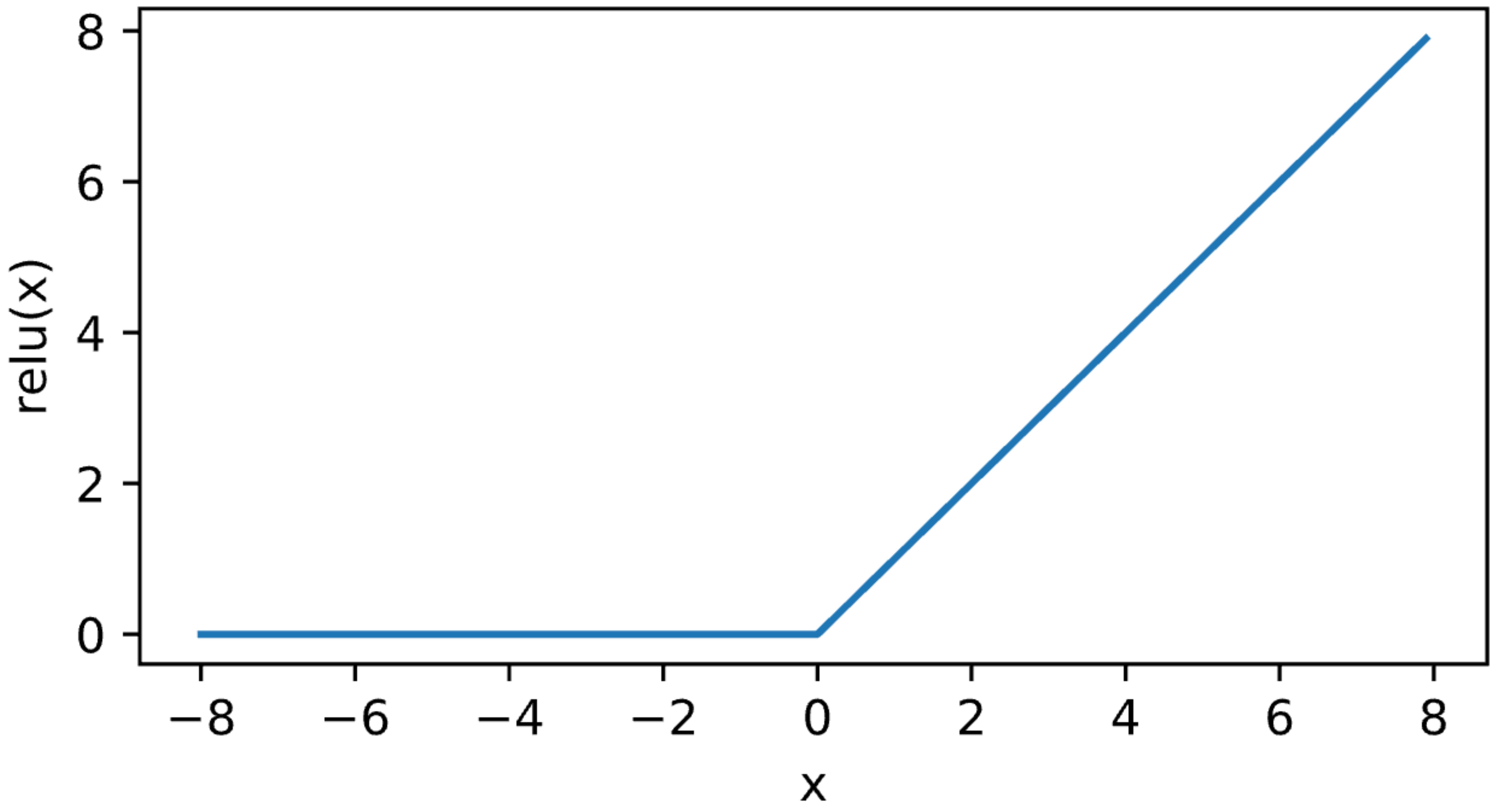


smooth version from [-1, 1]

ReLU Activation

ReLU: rectified linear unit (commonly used in modern neural networks)

$$\mathrm{ReLU}(x) = max(x, 0)$$



Quiz Break

Which one of the following is valid activation function

a)Step function
b) Sigmoid function
C) ReLU function
D) all of above

Quiz Break

Which one of the following is valid activation function

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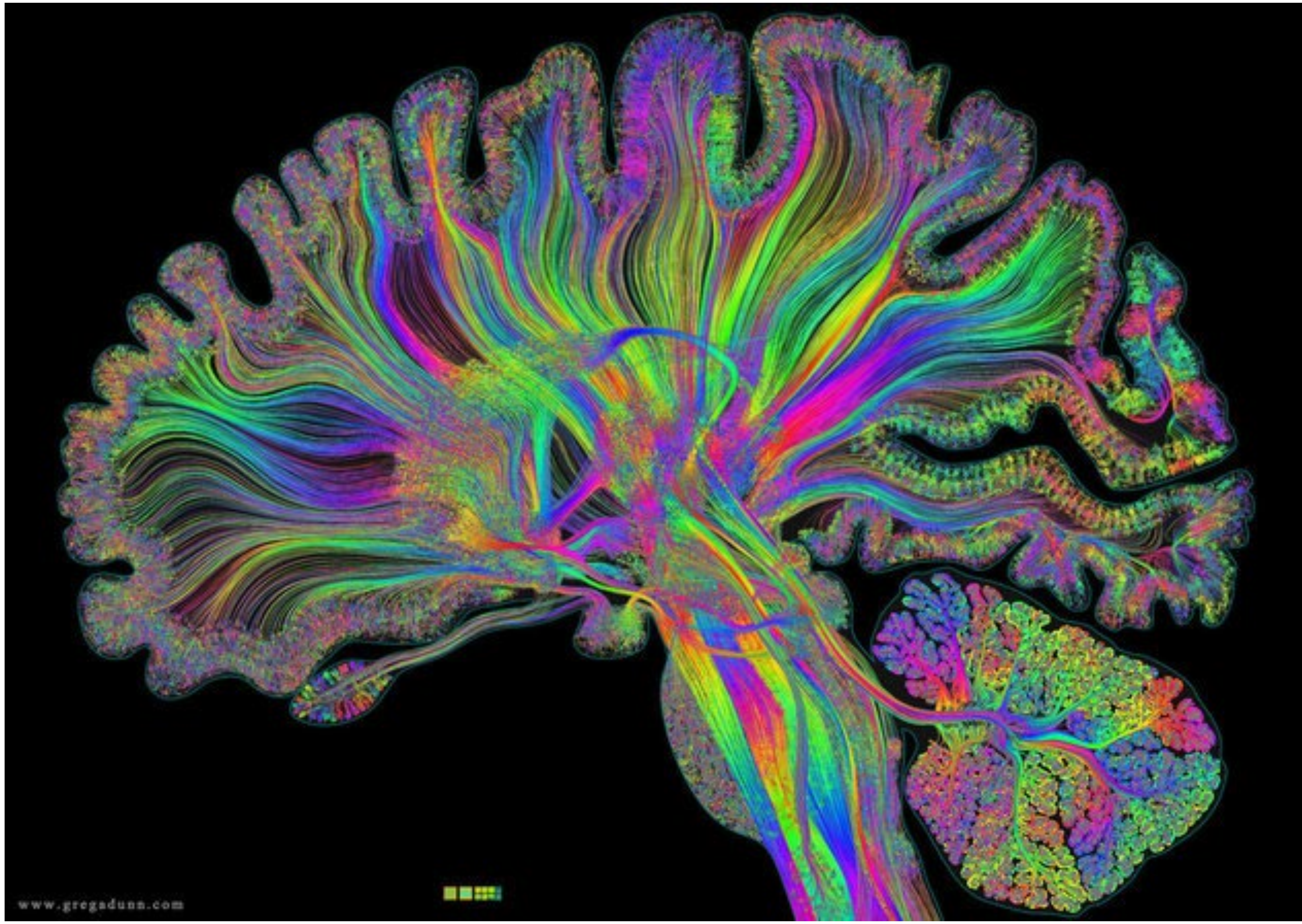
Multilayer Perceptron



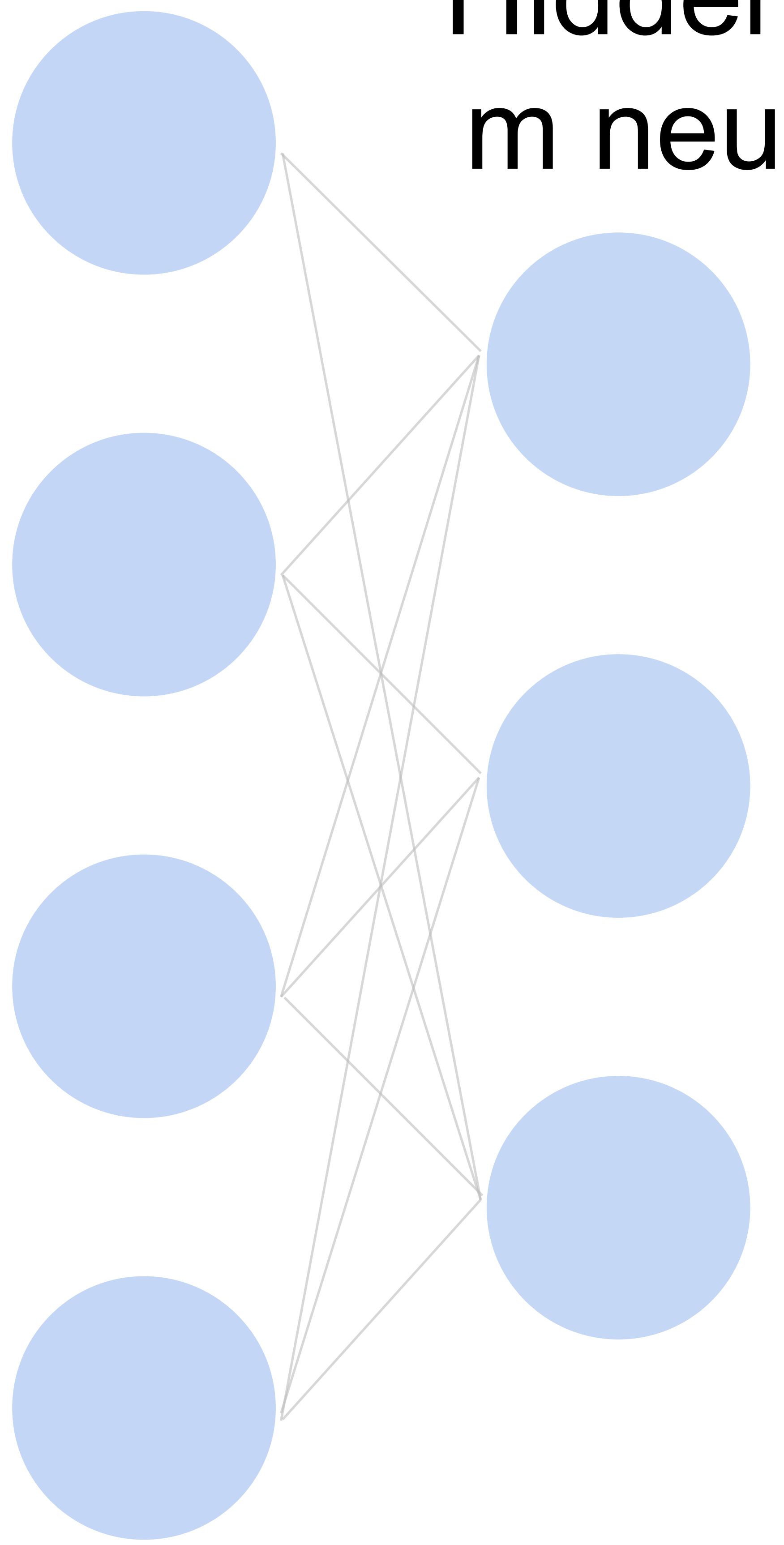
Single Hidden Layer

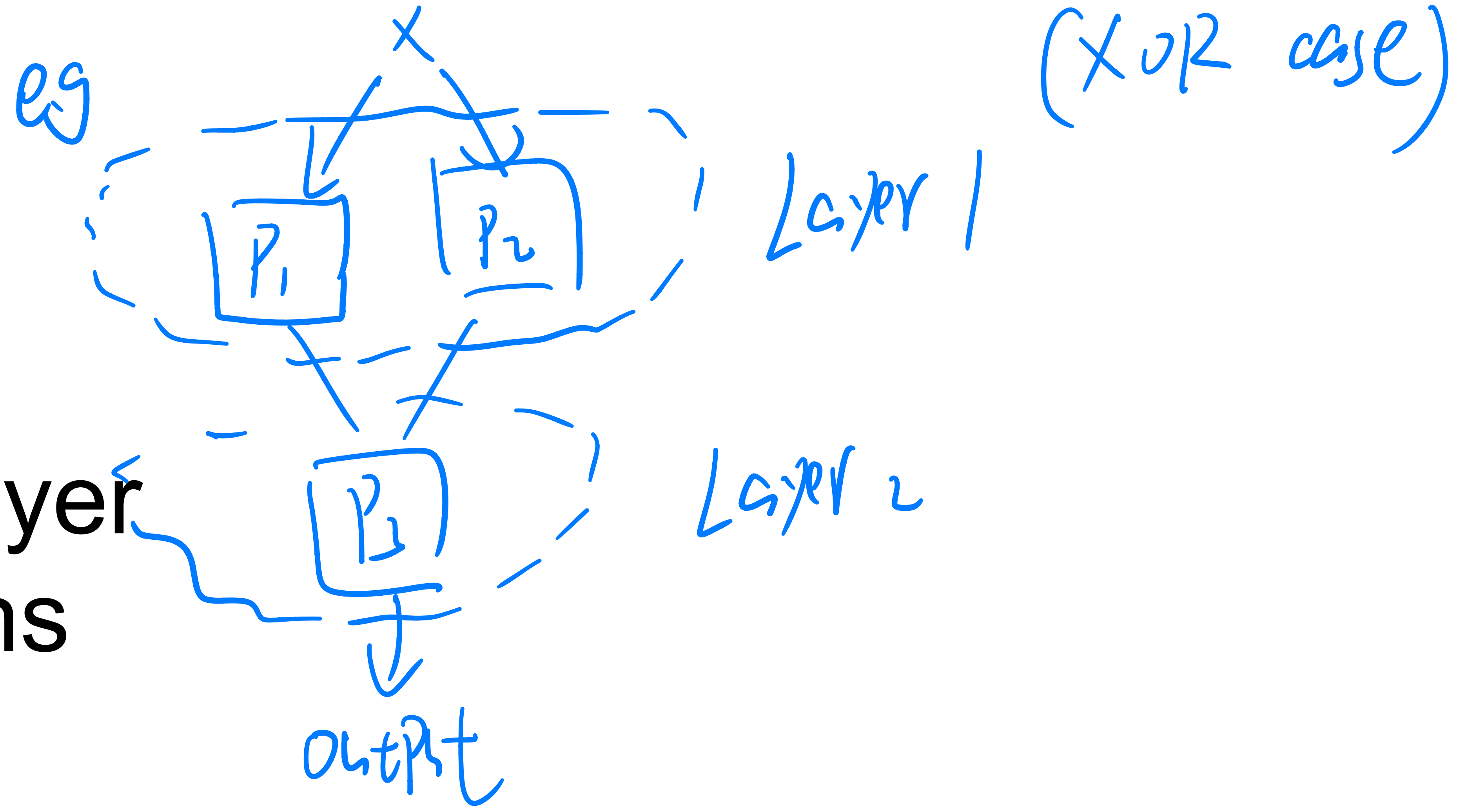
How to classify

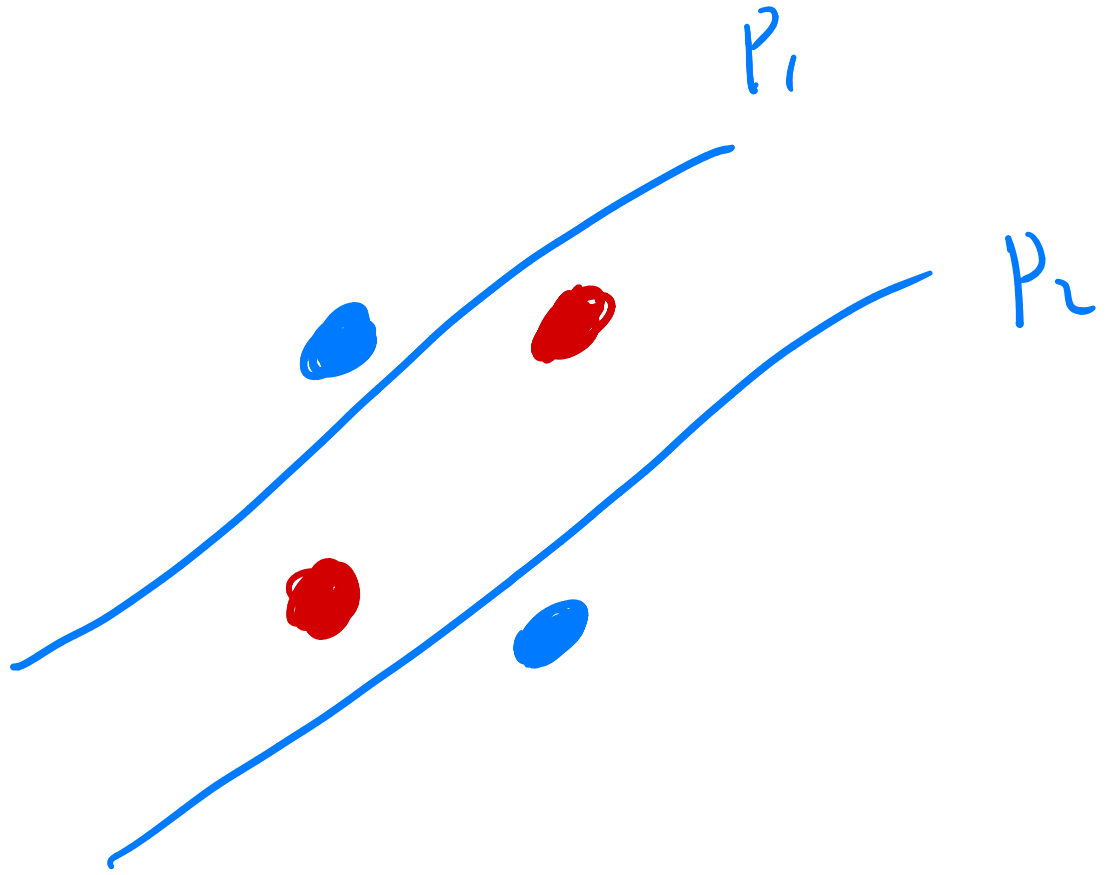
Cats vs. dogs?

Input

Hidden layer
m neurons



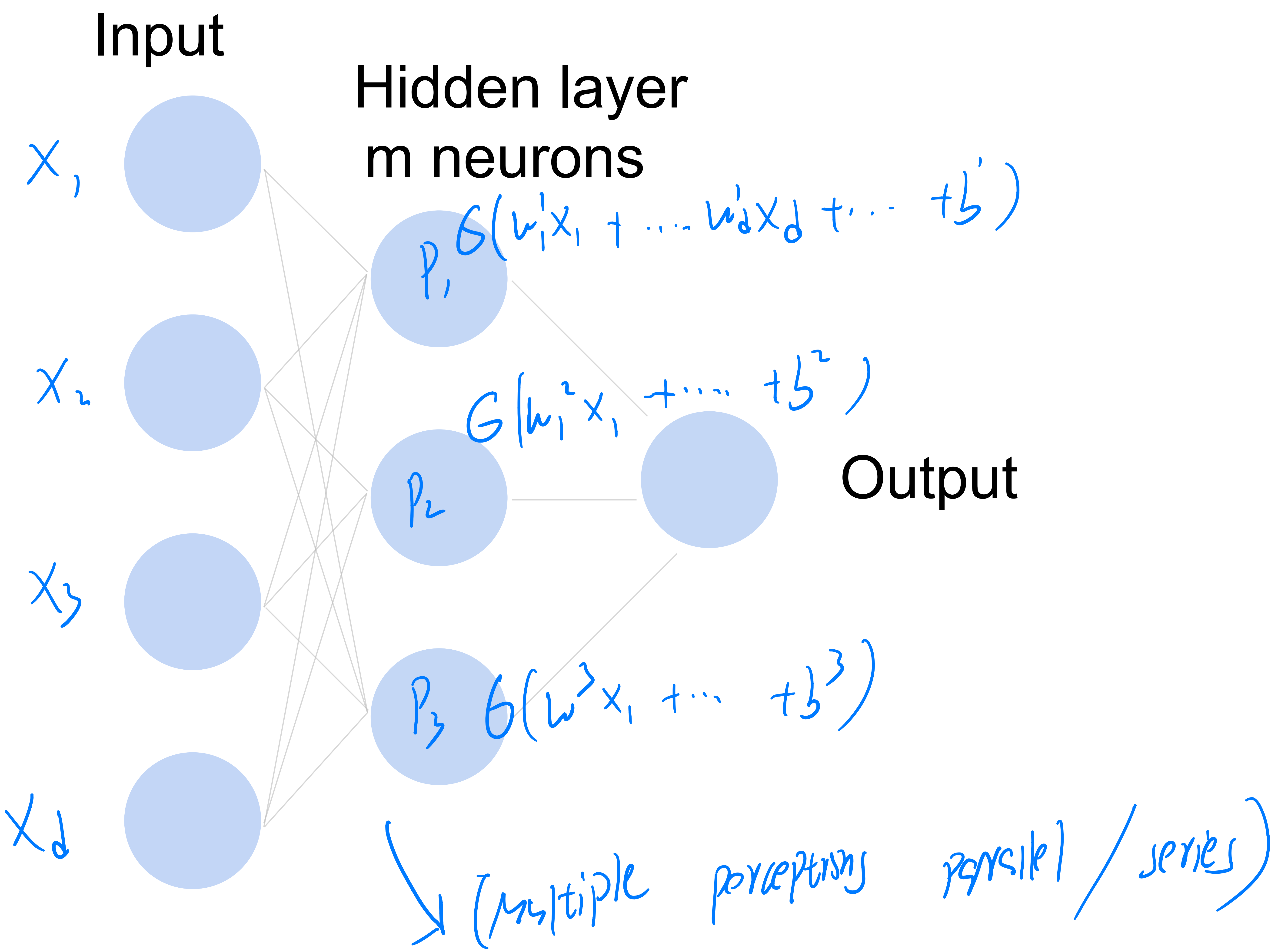




Single Hidden Layer

How to classify

Cats vs. dogs?



Single Hidden Layer

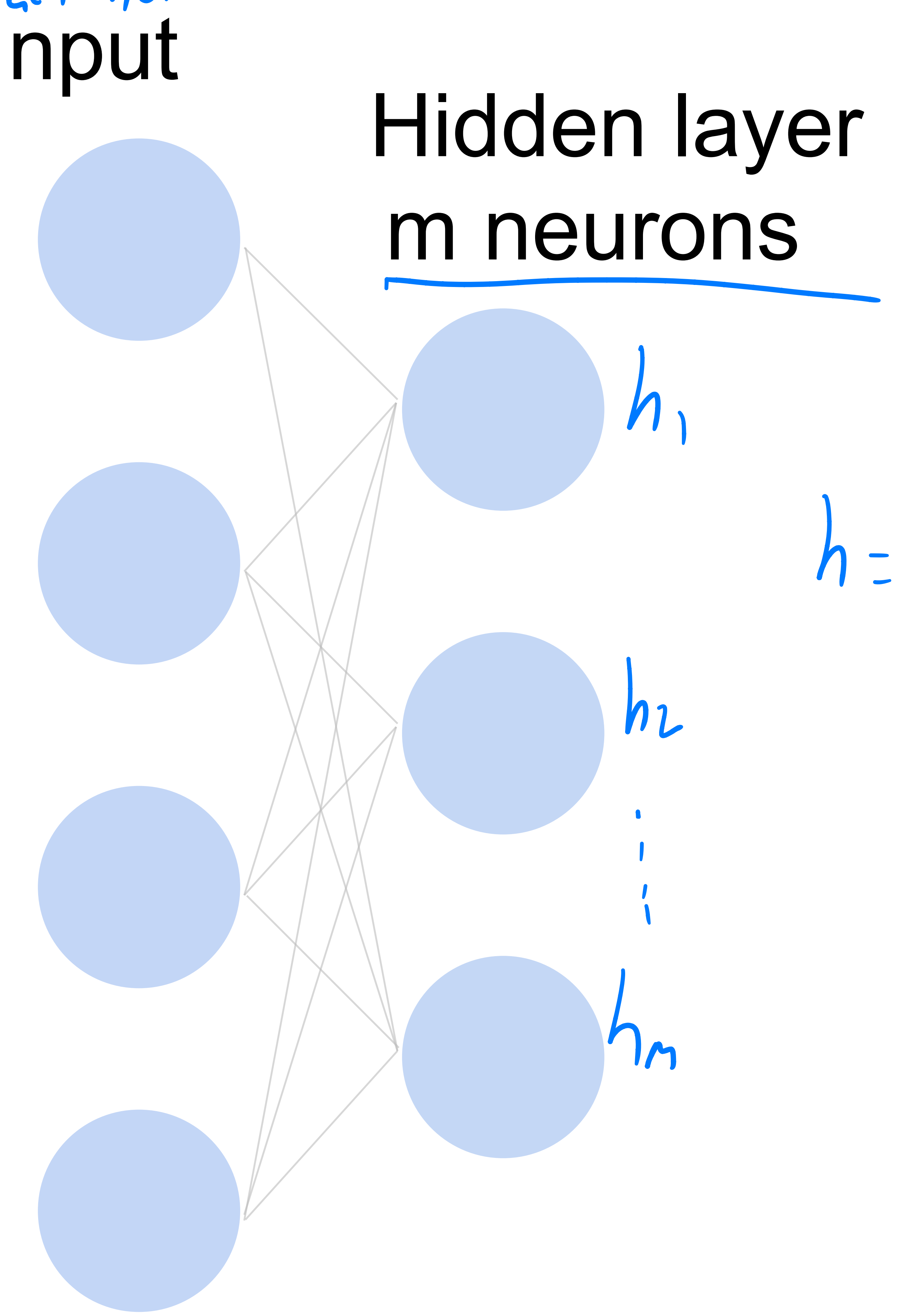
(m neurons, each neuron has d w_i corresponding to x_i)

- Input $\mathbf{x} \in \mathbb{R}^d$
- Hidden $\mathbf{W} \in \mathbb{R}^{m \times d}, \mathbf{b} \in \mathbb{R}^m$
- Intermediate output

$$\mathbf{h} \quad = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$$

Matrix-vector multiplication

σ is an element-wise activation function



$$h = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_m \end{bmatrix}$$

Quiz Break

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Which of the following functions is NOT an element-wise operation that can be used as an activation function?

A $\mathrm{f}(x) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

B $\mathrm{f}(x) = \begin{bmatrix} \max(0, x_1) \\ \max(0, x_2) \end{bmatrix}$

C $\mathrm{f}(x) = \begin{bmatrix} \exp(x_1) \\ \exp(x_2) \end{bmatrix}$

D $\mathrm{f}(x) = \begin{bmatrix} \exp(x_1 + x_2) \\ \exp(x_2) \end{bmatrix}$

Quiz Break

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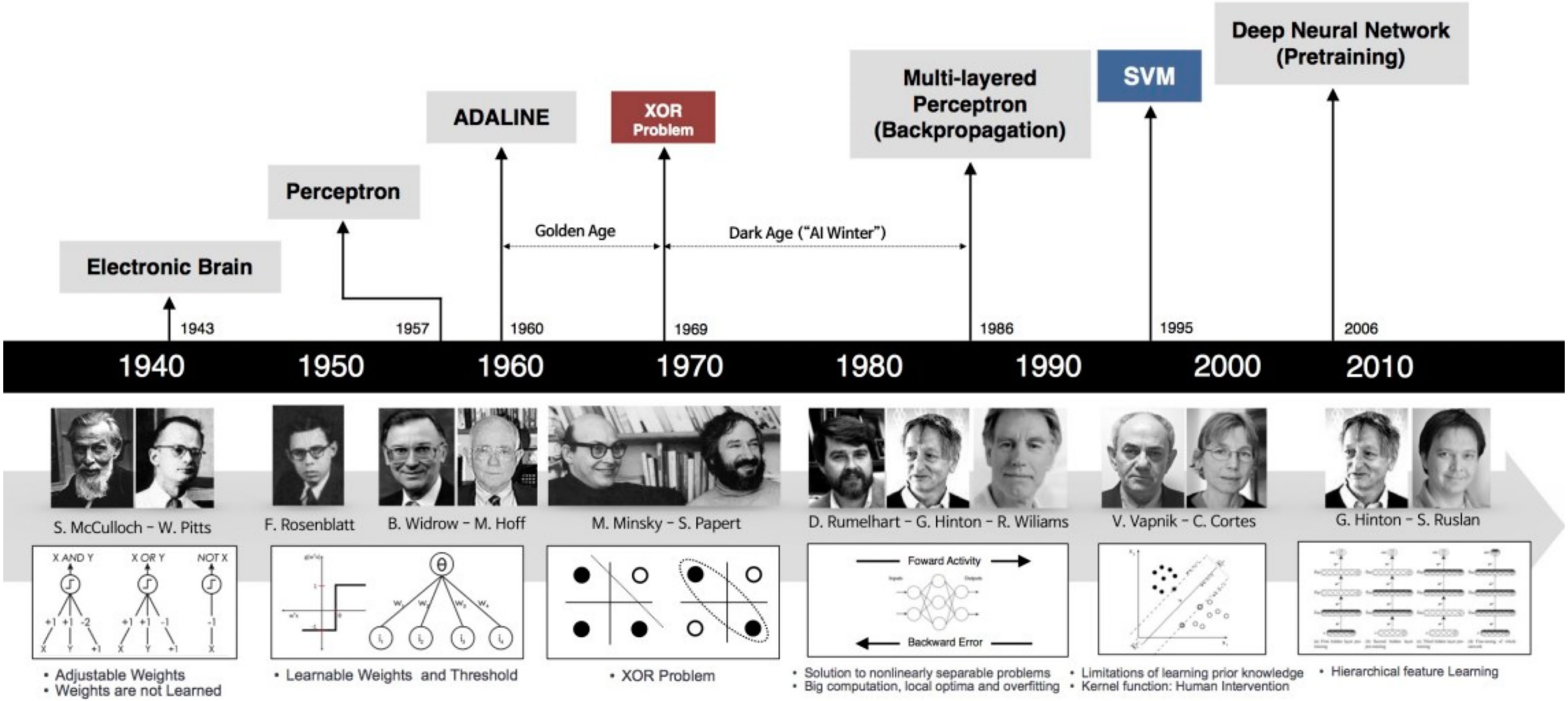
Brief history of neural networks



What we’ve learned today…

- Single-layer Perceptron
  - Motivation
  - Activation function
  - Representing AND, OR, NOT
- Brief history of neural networks